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CE : CIVIL ENGINEERING

Solutions on Topicwise Examination Questions

CONTENTS

Chapter No.	Topic	Page No.
1.	Engineering Mathematics	1
2.	Engineering Mechanics	53
3.	Solid Mechanics and Structural Analysis	55
4.	Concrete Structure	128
5.	Construction Materials and Management	145
6.	Steel Structure	154
7.	Soil Mechanics	180
8.	Fluid Mechanics and Hydraulics	253

Chapter No.	Topic	Page No.
9.	Environmental Engineering	**305**
10.	Irrigation and Hydrology	**339**
11.	Transportation Engineering	**369**
12.	Geomatics Engineering	**404**
•	General Aptitude (GA) Questions	**423**

2000

1. (B)

2. (B)

$$\lim_{a\to\infty}\int_1^a x^{-4}dn = \lim_{a\to\infty}\left[-\frac{1}{3}[x^{-3}]_1^a\right]$$

$$= \lim_{a\to\infty} -\frac{1}{3}\left[a^{-3}-1\right]$$

$$= -\frac{1}{3}[0-1] = \frac{1}{3}$$

3. (A)

$$f(x) = 1;\ -\frac{\pi}{2} < x < \frac{\pi}{2}$$

$$= 0;\text{ in other region}$$

$$\therefore\quad \frac{a_0}{2} = \frac{1}{2\pi}\int_{-\pi/2}^{\pi/2} 1dx = \frac{1}{2}$$

$$a_x = \frac{1}{\pi}\int_{-\pi/2}^{\pi/2} 1\cos x\,dx = \frac{2}{n\pi}\sin\frac{n\pi}{2}$$

$b_n = 0$ [$\because$ f(x) is even function]

Hence series is, $f(x) = \frac{1}{2} + \sum_{n=1}^{\infty}\frac{2}{n\pi}\sin\frac{nx}{2}\cos nx$

4. (B)

5. (A)

$$f = V = (x^2 + y^2 + z^2)^{-1/2}$$

$$\therefore\quad \frac{\partial V}{\partial x} = -x(x^2 + y^2 + z^2)^{-3/2}$$

or $\frac{\partial^2 V}{\partial y^2} = x(x^2 + y^2 + z^2)^{-5/2}(2x^2 - y^2 - z^2)$ ….(1)

Similarly

$\frac{\partial^2 V}{\partial x^2} = (x^2 + y^2 + z^2)^{-5/2}(2y^2 - x^2 - z^2)$ ….(2)

and $\frac{\partial^2 V}{\partial z^2} = (x^2 + y^2 + z^2)^{-5/2}(2z^2 - x^2 - y^2)$ ….(3)

Adding equations (1), (2) and (3), we get

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial z^2} = \frac{\partial^2 V}{\partial z^2} = 0$$

6. (A)

$$y_m = \sin\left(x + \frac{n\pi}{2}\right)$$

$$\therefore \quad f\left(\frac{\pi}{6}\right) = \sin\left(\frac{\pi}{6} + 0\right) = \frac{1}{2}$$

$$f'\left(\frac{\pi}{6}\right) = \sin\left(\frac{\pi}{6} + \frac{\pi}{2}\right) = -\frac{\sqrt{3}}{2}$$

$$f''\left(\frac{\pi}{6}\right) = \sin\left(\frac{\pi}{6} + \pi\right) = -\frac{1}{4}$$

Hence the required expression is

$$\sin x = \frac{1}{2} + \frac{\sqrt{3}}{2}\left(x - \frac{\pi}{6}\right) - \frac{1}{4}\left(x - \frac{\pi}{6}\right)^2 ....$$

7. (D)

8. (A)

$$f(x) = \lim_{x\to\infty}\left(1 - \frac{a^4}{x^4}\right) = \left(1 - \frac{a^4}{\infty}\right) = 1 - 0 = 1$$

9. (C)

$$f(x) = 2x^3 - 15x^2 + 36x + 10$$

$\therefore \quad f'(x) = 6x^2 - 30x + 36 \quad ....(1)$

and $\quad f''(x) = 12x - 30 \quad ....(2)$

For maxima and minima

$$f''(x) = 0$$

$\therefore \quad 6x^2 - 30x + 36 = 0$

or $\quad x^2 - 5x + 6 = 0$

or $\quad x = 3, 2$

Putting these values of x in equation (2); we get

$$f''(3) = 36 - 30 = +6$$

+ve, hence minimum value = 3

$$f''(2) = 24 - 30 = -6$$

–ve, hence maximum value = 2

10. Given $y(0) = 0$; $\frac{dy}{dt}(0) = 0$

$$(D^2 + \lambda^2)y = \cos \omega t + k$$

Then Auxilliary Equation (A.E.) is

$$m^2 + \lambda^2 = 0$$

$\Rightarrow \quad m = \pm \lambda i$

Hence C.F. = $C_1\cos\lambda t + C_2 \sin\lambda t \quad ....(1)$

For particular integral (P.I.)

$$\frac{1}{(D^2 + \lambda^2)} = (\cos \omega t + K.e^{0.t})$$

$$= \frac{\cos\omega t}{(\lambda^2 - w^2)} + \frac{k}{\lambda^2} \qquad \text{....(2)}$$

Hence general solution of equation is

$$y = c_1 \cos \lambda t + c_2 \sin \lambda t + \frac{\cos\omega t}{(\lambda^2 - w^2)} + \frac{k}{\lambda^2} \qquad \text{....(3)}$$

Then $\frac{dy}{dt} = \lambda c_1 \sin\lambda t + \lambda c_2 \cos \lambda t - \frac{\omega \sin\omega t}{\lambda^2 - \omega^2}$ ....(4)

Using boundary conditions

From equation (4), at t = 0; $\frac{dy}{dt} = 0$

or $$0 = \lambda C_1 + \frac{1}{(\lambda^2 - \omega^2)} + \frac{k}{\lambda^2}$$

or $$C_1 = -\left[\frac{k}{\lambda^2} + \frac{1}{(\lambda^2 - \omega^2)}\right]$$

Hence required expression is

$$y = -\left[\frac{k}{\lambda^2} + \frac{1}{(\lambda^2 - \omega^2)}\right] \cos\lambda t + \frac{\cos\omega t}{(\lambda^2 - \omega^2)} + \frac{k}{\lambda^2}$$

11. $$A = \begin{bmatrix} -2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

$$\therefore \quad |A - \lambda I| = \begin{bmatrix} 2-\lambda & -1 & 0 \\ -1 & 2-\lambda & -1 \\ 0 & -1 & 2-\lambda \end{bmatrix}$$

$$= (2 - \lambda)(\lambda^2 - 4\lambda + 2)$$
$$= -\lambda^3 + 6\lambda^2 - 10\lambda + 4$$

For eigen values

$$|A - \lambda I| = 0$$

or $-\lambda^3 + 6\lambda^2 - 10\lambda + 4 = 0$

or $(\lambda - 2)[(\lambda - 2)^2 - 2] = 0$

Hence $\lambda_1 = 2$

and $(\lambda - 2) = \pm\sqrt{2}$

or $\lambda_2 = 2 + \sqrt{2}$

and $\lambda_3 = 2 - \sqrt{2}$

The lowest eigen value is

$$\lambda_3 = 2 - \sqrt{2}$$

The eigen vector

$$\begin{bmatrix} \sqrt{2} & -1 & 0 \\ -1 & \sqrt{2} & -1 \\ 0 & -1 & \sqrt{2} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$\therefore \quad \sqrt{2}\,x - y = 0 \quad \ldots.(1)$

$-x + \sqrt{2}\,y - z = 0 \quad \ldots.(2)$

$-y + \sqrt{2}\,z = 0 \quad \ldots.(3)$

or $\quad \dfrac{x}{1} = \dfrac{y}{0} = \dfrac{z}{1}$

Hence the eigen vector is, $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^T$

Comparing from given type, we get
a = 0 and b = 1

2001

1. (A)

2. (A)

$$I = \int_0^{\pi/4} \cos^2 x \cdot dx = \int_0^{\pi/4} \frac{1+\cos 2x}{2}\, dx$$

$$= \int_0^{\pi/4} \frac{dx}{2} + \int_0^{\pi/4} \frac{\cos 2x}{2}\, dx$$

$$= \frac{1}{2}\,[x]_0^{\pi/4} + \frac{1}{2}\left[\frac{\sin 2x}{2}\right]_0^{\pi/4}$$

$$= \frac{\pi}{8} + \frac{1}{4}\left[\sin\frac{\pi}{2} - \sin x\right]$$

$$= \frac{\pi}{8} + \frac{1}{4} \times \frac{\pi}{8} + \frac{1}{4}$$

3. (D)

$$\underset{x\to\frac{\pi}{2}}{\text{Lt}}\ f(x) = \underset{x\to\frac{\pi}{2}}{\text{Lt}}\ \sin x = 1$$

4. (B)

$$\begin{bmatrix} 5 & 3 & 2 \\ 1 & 2 & 6 \\ 3 & 5 & 10 \end{bmatrix} = 5\,(20 - 30) - 3\,(10 - 18) + 2\,(5 - 6)$$

$$= -50 + 24 - 2 = -28$$

5. (D)

$$\frac{1}{S(S+2)} = \left[\frac{1}{S} - \frac{1}{S+2}\right] = \frac{1}{2}\left[1 - e^{-2t}\right]$$

6. (C)

$$\frac{dy}{dx} = \frac{3x^2}{2} - 2x + C_1$$

Integrating, we get,

$$y = \frac{3x^3}{6} - \frac{2x^2}{2} + C_1x + C_2$$

$\therefore$ $y(0) = 2 = 0 + 10 + C_2$

or $C_2 = 2$

Again $-3 = \frac{3}{2}(1)^2 - 2(1) + C_{11}$

or $-3 = 1.5 - 2 + C_1$

or $C_1 = -3 + 0.5 = -2.5 = -\frac{5}{2}$

$$\therefore \quad y = \frac{x^3}{2} - x^2 - \frac{5}{2}x + 2$$

7. (A)

$$[P]\,[Q]^T = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}\begin{bmatrix} 4 & 9 \\ 8 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 8+24 & 18+6 \\ 16+40 & 56+10 \end{bmatrix}$$

$$= \begin{bmatrix} 32 & 24 \\ 56 & 46 \end{bmatrix}$$

8. (D)

$$\begin{vmatrix} 5-\lambda & 3 \\ 2 & 9-\lambda \end{vmatrix} = 0$$

or $(5 - \lambda)(9 - \lambda) - 6 = 0$

or $45 - 9\lambda - 5\lambda + \lambda^2 - 6 = 0$

or $\lambda - 14\lambda + 39 = 0$

Solving we get, $\lambda = 10.1623, 3.837$

Hence eigen value of the matrix are (10.16, 3.837)

9. Let the velocity of engine in time t is V. Then

$F \propto V^2$

or $F = KV^2$

$F = m_a = KV^2$

$\therefore$ $a = \frac{K}{m}V^2$

or $a = V\frac{dV}{dx} = \frac{K}{m}V^2$

or $\frac{dV}{dx} = \frac{K}{m}V$

$\therefore \quad \frac{m}{K} \cdot \frac{dV}{V} = dx$

10. Let side of parallelepiped is 2x, 2y, 2z

$\therefore \quad V = 8\,xyz$

Equation of hemisphere for positive 'z–axis is

$z = \sqrt{a^2 - x^2 - y^2}$

$\therefore \quad V = 8xy \cdot \sqrt{a^2 - x^2 - y^2}$

Differentiating, we get

$$\frac{\partial V}{\partial y} = 8y \cdot \sqrt{a^2 - x^2 - y^2} + \frac{8xy \cdot (-2x)}{2\sqrt{a^2 - x^2 - y^2}} = 0$$

and

$$\frac{\partial V}{\partial y} = 8x \cdot \sqrt{a^2 - x^2 - y^2} + \frac{8xy(-2y)}{2\sqrt{a^2 - x^2 - y^2}} = 0$$

From these equations, x = y

Again equation of hemisphere for positive y – axis

$y = \sqrt{a^2 - x^2 - y^2 - z^2}$

Again partially differentiating and y = 2

$\therefore \quad x = y = 2$

or $\quad z = \sqrt{a^2 - x^2 - y^2}$

or $\quad x^2 + y^2 + z^2 = a^2$

$\therefore \quad x = y = z = \frac{a}{\sqrt{3}}$

Hence the volume is, $V = 8 \times \frac{a}{\sqrt{3}} \times \frac{a}{\sqrt{3}} \times \frac{a}{\sqrt{3}} = \frac{8}{3\sqrt{3}}\, a^3$

2002

1. (A)

2. (D)

3. (C)

4. (C)

5. (A)

6. (C)

7. (B)

8. The volume of the solid bounded by coordinate planes and the plane x + y + z = 2 is

$$= \int_0^2 \int_0^{2-x} (2 - x - y)\, dx\, dy$$

But mass of solid = Volume × Density

$$= \int_0^2 \int_0^{2-x} (2 - x - y)\, dx\, dy \times 2x$$

$$= \int_0^2 \int_0^{2-x} (4x - 2x^2 - 2xy)\, dydx$$

$$= \int_0^2 \left\{ 4x(y)_0^{2-x} - 2x^2(y)_0^{2-x} - 2x\left(\frac{y^2}{2}\right)_0^{2-x} \right\} dx$$

$$= \int_0^2 \left\{ 4x(2-x) - 2x^2(2-x) - x(2-x)^2 \right\} dx$$

$$= \int_0^2 \left\{ 8x - 4x^2 - 4x^2 + 2x^3 - x(4 + x^2 - 2x) \right\} dx$$

$$= \int_0^2 (8x - 4x^2 - 4x^2 + 2x^3 - 4x - x^3 + 2x^2)\, dx$$

$$= 4.\left[\frac{x^2}{2}\right]_0^2 - 6.\left[\frac{x^3}{3}\right]^2 + \left[\frac{x^4}{4}\right]_0^2$$

$$= 2.[x - 0] - 2[8 - 0] + \frac{1}{4}[16 - 0]$$

= 8 – 16 + 4

= – 4 i.e. 4 unit

9. Given equation is

$$\frac{dN}{dt} = -aN\left(\ln\frac{N}{K}\right)$$

$$\therefore \quad \frac{dN}{N\left(\ln\frac{N}{K}\right)} = -adt$$

On integration we get,

$$\ln\left\{\ln\frac{N_0}{K}\right\} = -at + C \qquad \text{....(i)}$$

Given at t = 0; N = N_0

$$\therefore \quad \ln\left\{\ln\frac{N}{K}\right\} = C$$

Using the volume of C in equation (i)

$$\ln\left\{\ln\frac{N}{K}\right\} = -at + \ln\left\{\ln\frac{N_0}{K}\right\}$$

$$\Rightarrow \quad \ln\left\{\frac{\ln N/K}{\ln N_0/K}\right\} = -at$$

$$\Rightarrow \quad \ln\frac{N}{K} = e^{-at}\ln\frac{N_0}{K} \qquad \text{....(ii)}$$

(a) Equation (ii) is the required equation.

(b) Again, $\ln\frac{N}{K} = \underset{t\to\infty}{Lt}\, e^{-at}.\ln\frac{N_0}{K} = 0$

$$\Rightarrow \quad \frac{N}{K} = e^0$$

So, $N = K$ i.e. required population.

(c) Since, grow rate is maximum at $N = eK$

$$\therefore \quad \ln\frac{CK}{K} = 0$$

$$\Rightarrow \quad \ln C = 0$$

$$\Rightarrow \quad e = 1 \,\varepsilon\, [0, 1]$$

2003

1. (C)

Given matrix $[A] = \begin{bmatrix} 4 & 2 & 1 & 3 \\ 6 & 3 & 4 & 7 \\ 2 & 1 & 0 & 1 \end{bmatrix}$

Eliminating first column and expanding the determinant of the given matrix, we get

$$|A| = \begin{bmatrix} 4 & 2 & 1 & 3 \\ 6 & 3 & 4 & 7 \\ 2 & 1 & 0 & 1 \end{bmatrix} = 0$$

Therefore r(A) is less than 3.

We observe that $\begin{bmatrix} 4 & 7 \\ 0 & 1 \end{bmatrix}$ is a non-singular square sub-matrix of order 2.

Hence $r(A) = 2$

By definition, the rank of a matrix is of the highest order non-singular square sub-matrix.

2. (D)

3. (A)

The general equation of a plane passing through three points will be

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$$

Since it passing through (0, 0, 0), therefore equation becomes,

$$ax + by + cz = 0 \qquad \text{....(i)}$$

Again it is passes through, (1, 3, 4) and (2, 1, –2), therefore

$$a + 3b + 4c = 0$$

and $2a + b - 2c = 0$

By cross multiplication, we get

$$\frac{a}{-6-4} = \frac{b}{+8+2} = \frac{c}{+1-6} = \lambda \text{ (say)}$$

$\therefore$ $a = -10\lambda$

$b = +10\lambda$

$c = -5\lambda$

Putting these values in equation (i), we get

$-10\lambda x + 10\lambda y - 5\lambda z = 0$

or $-10x + 10y - 5z = 0$

or $10x - 10y + 5z = 0$

or $2x - 2y + z = 0$

Perpendicular distance of this plane from points (3, –2, –1) will be,

$$\left|\frac{6+4-1}{\sqrt{4+4+1}}\right| = \left|\frac{9}{3}\right| = 3$$

4. (B)

By definition of Laplace transforms

$$L\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt$$

$$L[\sin(at)] = \frac{a}{s^2+a^2}$$

$$L[\cos(at)] = \frac{S}{a^2+s^2}$$

5. (B)

According to Fourier series,

$$1 + \frac{2x}{\pi} = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

where, $a_0 = \frac{1}{\pi}\int_{-\pi}^{0} f(x)\,dx = \frac{1}{\pi}\int_{-\pi}^{0}\left[1+\frac{2x}{\pi}\right]dx$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{0} f(x)\cos nx\,dx$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{0} f(x)\sin nx\,dx$$

Again, let

$$\left(1+\frac{2x}{\pi}\right) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

where, $a_0 = \frac{1}{\pi}\int_0^{\pi} f(x)\,dx$

$$a_n = \frac{1}{\pi}\int_0^{\pi} f(x)\cos nx\,dx$$

$$b_n = \frac{1}{\pi}\int_0^{\pi} f(x)\sin nx\,dx$$

Adding all these terms, we get

$$\sum_{n=1}^{\infty}\frac{4}{\pi^2 n^2}(1-\cos nx)$$

2004

1. (A)

2. (D)

Given: $S = 2 + \frac{5}{2}+\frac{8}{2^2}+\frac{11}{2^3} + \ldots \infty$ …(i)

So, $\frac{1}{2}S = \frac{1}{2}\times 2 + \frac{5}{2^2} + \frac{8}{2^3} + \frac{11}{2^4} + \ldots \infty$ …(ii)

Subtracting equation (ii) from equation (i), we get

$$S - \frac{1}{2}S = 2 + \frac{5-2}{2} + \frac{8-5}{2^2} + \frac{1-8}{2^3} + \ldots\ldots$$

$$\frac{S}{2} = 2 + \frac{3}{2} + \frac{3}{2^2} + \frac{3}{2^3} + \ldots\ldots$$

$$= 2 + 3\left(\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}\right) = 2 + 3\left(\frac{\frac{1}{2}}{1-\frac{1}{2}}\right) = 5$$

$S = 10$

3. (B)

$$f(x) = \lim_{x\to 0}\frac{x^2+x^2}{2x^3-7x^2}$$

Applying L. Hospital rule

[Put $x \to 0$, then $\frac{0}{0}$, $\frac{\infty}{\infty}$ + …]

$$f'(x) = \frac{3x^2-2x}{6x^2-14x}$$ [again $\frac{0}{0}$ form]

$$f''(x) = \frac{6x+2}{12x-14}$$

then $f''(x) = \lim_{x\to 0}\frac{6(0)+2}{12(0)-14} = \frac{2}{-14} = \frac{-1}{7}$

4. (C)

$$\begin{vmatrix} 4-\lambda & -2 \\ -2 & 1-\lambda \end{vmatrix} = 0$$

$\Rightarrow (4 - \lambda)(1 - \lambda) - 4 = 0$

$\Rightarrow 4 + \lambda^2 - 5\lambda - 4 = 0$

$\Rightarrow \lambda^2 - 5\lambda = 0$

$\Rightarrow \lambda(\lambda - 5) = 0$

$\Rightarrow \lambda = 0, 5$

5. (A)

For maximum f(x), $\dfrac{d[f(x)]}{dx} = 0$

$\Rightarrow$ $f'(x) = 0$

$\Rightarrow$ $6x^2 - 6x - 36 = 0$

$\Rightarrow$ $x^2 - x - 6 = 0$

$\Rightarrow$ $x^2 - 3x + 2x - 6 = 0$

$\Rightarrow$ $x(x - 3) + 2(x - 3) = 0$

$\Rightarrow$ $x = 3, -2$

$\Rightarrow$ $f''(x) = 12x - 6$

So, f(x) has man value at -2 only.

6. (B)

Given: $\dfrac{dx}{dt} + kx^2 = 0$

$\Rightarrow$ $\dfrac{dx}{dt} = -kx^2$

Integrating, we have $\int \dfrac{dx}{x^2} = \int -kdt \left[\therefore \dfrac{x^{-2+1}}{-2+1} = -kt\right]$

$\Rightarrow$ $-\dfrac{1}{x} = -kt + C$

If x = a, t = 0

$\therefore$ $C = -\dfrac{1}{a}$

$\therefore$ $-\dfrac{1}{x} = -kt - \dfrac{1}{a}$

$\Rightarrow$ $\dfrac{1}{x} = \dfrac{1}{a} + kt$

7. (C)

1	2	3	4
failed	f	f	not failed
+			
f	f	f	f

$\therefore$ Probability $= 1 \times 0.2 \times 0.2 \times 0.8 + 1 \times 0.2 \times 0.2 \times 0.2 = 0.032 + 0.008 = 0.04$

2005

1. (A)

2. (A)

3. (C)

4. (D)

5. (D)

6. (A)

$$\frac{dy}{dt} + p(t)y = q(t)y^n,\ n > 0$$

Dividing the both sides by y^n, we get

$$y^{-n}\frac{dy}{dt} + p(t)\,y^{1-n} = p(t) \qquad \text{....(i)}$$

Let $v = y^{1-n}$, then $\frac{dy}{dt} = \left[(1-n)\,y^{-n}\frac{dy}{dt}\right]$

Putting these values in equation (i), we get

$$\frac{1}{1-n}\cdot\frac{dv}{dt} + p(t)v = q(t)$$

Multiply both sides by (1 – n), we get

$$\frac{dv}{dt} + (1-n)\,p(t)\,v = q(t)\ (1 - v)$$

7. (D)

According to mean value theorem, the speedometer at a centre line during each reading

$$= \frac{\text{Travelled distance}}{\text{Time}} = \frac{280}{8}\ \text{m/sec}$$

$$= \frac{280}{8} \times \frac{18}{5} = 128\ \text{km/h}$$

8. (A)

Given, $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 17y = 0$

$\therefore \quad (D^2 + 2D + 17)y = 0$

$\Rightarrow \quad D^2 + 2D + 17 = 0$

$\therefore \quad D = -1 \pm 4i$

Hence, its solution is $y = e^{-x}\left(c_1 e^{+i4x} + c_2 e^{-4ix}\right)$

$= e^{-x}\left\{c_1(\cos 4x + i\sin 4x) + c_2(\cos 4x - i\sin 4x)\right\}$

$= e^{-x}\left\{(c_1 + c_2)\cos 4x + i(c_1 - c_2)\sin 4x\right\}$ ....(i)

From given condition; $y(0) = 1$

$1 = c_1 + c_2$

Now, $\dfrac{dy}{dx} = e^{-x}\left\{-4(c_1+c_2)\sin 4x + 4i(c_1-c_2)\cos 4x\right\}$

$-e^{x}\left\{(c_1+c_2).\cos 4x + i(c_1-c_2)\cos 4x\right\}$ ….(ii)

From given condition, $\dfrac{dy}{dx}\left(\dfrac{\pi}{4}\right) = 0$

Putting this value in equation (ii), we get

$$0 = e^{-\pi/4}\left[4i(c_1-c_2)(-1)+(c_1+c_2)\right]$$

$\Rightarrow$ $c_1 + c_2 - 4i(c_1 - c_2) = 0$

Putting $c_1 + c_2 = 1$ from (i) condition, we get

$\therefore$ $i(c_1 - c_2) = \dfrac{1}{4}$

Putting the values of $c_1 + c_2$ and $i(c_1 - c_2)$ in equation (ii), we get solution of the differential equation as

$$y = e^{-x}\left(\cos 4x + \frac{1}{4}\sin 4x\right)$$

9. (B)

10. (A)

11. (C)

12. (B)

2006

1. (D)

Given, the set of equations,

$0.x + 4.y + 3.z = 8$

$2.x + 0.y - 1.z = 2$

$3.x + 2.y + 0.z = 5$

$$\begin{bmatrix} 0 & 4 & 3 \\ 2 & 0 & -1 \\ 3 & 2 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \\ 5 \end{bmatrix}$$

AX = B, where B = Argument matrix of A.

Applying $R_3 \to R_3 - R_2$, we get

$$\begin{bmatrix} 0 & 4 & 3 \\ 2 & 0 & -1 \\ -3 & 2 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \\ 5 \end{bmatrix}$$

Applying $R_2 \leftrightarrow R_1$, $R_2 \leftrightarrow R_3$, we get

$$\begin{bmatrix} 2 & 0 & -1 \\ 3 & 2 & 0 \\ 0 & 4 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix}$$

Applying $R_2 \to R_2 - \frac{3R_1}{2}$, we get

$$\begin{bmatrix} 2 & 0 & -1 \\ 0 & 2 & \frac{3}{2} \\ 0 & 4 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 8 \end{bmatrix}$$

Applying $R_3 \to R_3 - 2R_2$, we get

$$\begin{bmatrix} 2 & 0 & -1 \\ 0 & 2 & \frac{3}{2} \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}$$

$\therefore$ Rank of A $\neq$ Rank of B

As inconsistent, thus there is no solution.

2. (C)

3. (B)

Given, $A = \begin{bmatrix} 2 & -2 & 3 \\ -2 & -1 & 6 \\ 1 & 2 & 0 \end{bmatrix}$

Since $|A - \lambda I| = 0$,

$$\therefore \begin{vmatrix} 2-\lambda & -2 & 3 \\ -2 & -1-\lambda & 6 \\ 1 & 2 & -\lambda \end{vmatrix} = 0$$

$\Rightarrow \lambda^3 - \lambda^2 - 21\lambda + 45 = 0$

$\lambda_1 + \lambda_2 + \lambda_3 = -\left(-\frac{1}{1}\right) = 1$ ….(i)

$\lambda_1\lambda_2\lambda_3 = -45$ ….(ii)

Given $\lambda_1 = 3$,

$\therefore \lambda_2 + \lambda_3 = -2$

and $\lambda_2\lambda_3 = -15$

Solving, we get $\lambda_2 = 3$, $\lambda_3 = -5$

4. (C)

$f(x,y,z) = 2x^2 + 3y^2 + z^2$

$$\text{grad } f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k}$$

$$= 4x\hat{i} + 6y\frac{Pl^3}{3EI} + 2z\hat{k}$$

grad f at point (2, 1, 3) = 8i + 6j + 6k

$\therefore$ Unit vector in direction, $\hat{a} = \frac{i - 2k}{\sqrt{1+4}} = \frac{i - 2k}{\sqrt{5}}$

$\therefore$ Directional derivative = grad . $\hat{a}$

$$= \left(8\hat{i}+6\hat{j}+6\hat{k}\right)\cdot\left(\frac{i-2\hat{k}}{\sqrt{5}}\right)$$

$$= \frac{8}{\sqrt{2}}-\frac{12}{\sqrt{5}} = -1.789$$

5. (D)

6. (A)

7. (D)

At $r = a$, $x^2 + y^2 = a^2$

At $r = 2a\cos\theta$, $x^2 + y^2 - 2ax = 0$

$(x-a)^2 + y^2 = a^2$

Required area

$$= 2\left[\int_0^{a/2}\sqrt{a^2-(x-a)^2}\,dx \int_{a/2}^{a}\sqrt{a^2-x^2}\,dx\right]$$

$$= 2\left[\frac{\pi}{3}-\frac{\sqrt{3}}{4}\right]a^2 = \left(\frac{2\pi}{3}-\frac{\sqrt{3}}{2}\right)a^2$$

8. (A)

From Cauchy's integral theorem

$$f^n(a) = \frac{n!}{2pi}\int\frac{f(z)}{(z-a)^{n+1}} \qquad \text{....(i)}$$

Given, $\oint\frac{z^3-6}{3z-i} = \frac{1}{3}\oint\frac{z^3-6}{z-i/3}$

$f(z) = z^3 - 6$

$a = i/3$

$n = 0$

Comparing with equation (i), we get

$$f(i/3) = \frac{n!}{2pi}\oint\frac{z^3-6}{(z-i/3)}dz$$

$$\Rightarrow \frac{f(i/3)2pi}{0!} = \oint\frac{z^3-6}{z-i/3}$$

$$\Rightarrow \frac{\left[(i/3)^3-6\right]2\pi i}{i} = \oint\frac{z^3-6}{z-i/3}$$

$$\Rightarrow \left(\frac{i^3}{27}-6\right)2\pi i = 3\oint\frac{z^3-6}{3z-i}dz$$

$$\Rightarrow \frac{1}{3}\left(\frac{-i}{27}-6\right)2\pi i = \oint\frac{z^3-6}{3z-i}dz$$

$$\Rightarrow \frac{1}{3}\left(\frac{2\pi\times i^2}{27}-12\pi i\right) = \oint\frac{z^3-6}{3z-i}dz$$

$$\Rightarrow \quad \frac{2\pi}{81} - 4\pi i = \oint \frac{z^2 - 6}{3z - 1}$$

9. (B)

Required probability = $\dfrac{{}^{23}C_4 \cdot {}^{2}C_1}{{}^{25}C_5} = \dfrac{1}{3}$

10. (A)

11. (A)

Given, differential equation is

$$\frac{dy}{dx} + \left(\frac{2}{3}\right)y = \frac{1}{x} - \frac{1}{x^2}$$

$\therefore$ Integrating factor (I. F.) = $e^{\int\left(\frac{2}{x}\right)dx} = x^2$

Its solution is, $x^2y = \int x^2\left(\dfrac{1}{x} - \dfrac{1}{x^2}\right)dx$

$$\Rightarrow \quad x^2y = \int (x - 1)\,dx$$

$$\Rightarrow \quad x^2y = \frac{x^2}{2} - x + c$$

From end condition, $c = \dfrac{1}{2}$

$$\therefore \quad x^2y = \frac{x^2}{2} - x + \frac{1}{2}$$

$$\therefore \quad y = \frac{1}{2} - \frac{1}{x} + \frac{1}{2x^2}$$

2007

1. (B)

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

Eigen values are given by $|A - \lambda I| = 0$

$$\Rightarrow \quad \begin{bmatrix} 1-\lambda & 1 & 3 \\ 1 & 5-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{bmatrix} = 0$$

Apply $C_1 \longrightarrow C_1 - C_3$

$$\begin{bmatrix} -2-\lambda & 1 & 3 \\ 0 & 5-\lambda & 1 \\ 2+\lambda & 1 & 1-\lambda \end{bmatrix} = 0$$

$$\Rightarrow \quad (2+\lambda)\begin{vmatrix} -1 & 1 & 3 \\ 0 & 5-\lambda & 1 \\ 1 & 1 & 1-\lambda \end{vmatrix} = 0$$

$\Rightarrow$ $(2 + \lambda)$ $[-1\{(5 - \lambda) - 1\} + 1\{1 - 3(5 - \lambda)\}] = 0$

$\Rightarrow$ $(2 + \lambda)$ $[5 - 6\lambda + \lambda^2 - 1 - 1 + 15 - 3\lambda] = 0$

$\Rightarrow$ $(2 + \lambda)$ $(\lambda^2 - 9\lambda + 18) = 0$

$\Rightarrow$ $(2 + \lambda)$ $(\lambda - 6)$ $(\lambda - 3) = 0$

$\Rightarrow$ $\lambda = -2, 3, 6$

2. (B)

Degree of differential equation $\frac{d^2x}{dt^2} + 2x^3 = 0$ is 1

3. (D)

$$\frac{dy}{dx} = x^2y$$

Integrating $\int \frac{dy}{y} = \int x^2 dx$

$\Rightarrow$ $\log y = \frac{x^3}{3} C$

At $x = 0$, $y = 1$ $\Rightarrow$ $c = 0$

$\therefore$ $\log y = \frac{x^3}{3}$

$\Rightarrow$ $y = e^{x^3/3}$

4. (A)

Given equation can be written as, $AX = K$

where, $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & 3 \\ 1 & 2 & \alpha \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $K = \begin{bmatrix} 5 \\ 9 \\ \beta \end{bmatrix}$

For infinite solutions, = 0, $\Delta_x = 0$, $\Delta_y = 0$

and $\Delta_z = 0$

$$\therefore \quad \Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 3 & 3 \\ 1 & 2 & \alpha \end{vmatrix} = 0$$

$\Rightarrow$ $3\alpha - 6 - \alpha + 3 + 2 - 3 = 0$

$\Rightarrow$ $\alpha = 2$

$$\Delta_z = \begin{vmatrix} 1 & 1 & 5 \\ 1 & 3 & 9 \\ 1 & 2 & \beta \end{vmatrix} = 0$$

$\Rightarrow$ $(3\beta - 18) - 1(\beta - 9) + 5(2 - 3) = 0$

$\Rightarrow$ $2\beta - 14 = 0$

$\Rightarrow$ $\beta = 7$

5. (D)

Div. $\vec{V} = \frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}$

$\vec{V} = 5xy\,\vec{i} + 2y^2\,\vec{j} + 3yz^2\,\vec{k}$

$\frac{\partial V}{\partial x}\hat{i} = 5y + 0 + 0 = 5y$

$\frac{\partial v}{\partial y}\hat{j} = O + 4y + O = 4y$

$\frac{\partial V}{\partial z}\hat{k} = 0 + 0 + 6yz = 6yz$

div. $\vec{V} = 5y + 4y + 6yz$

At (1, 1, 1), div. $\underline{\vec{V}} = 5 + 4 + 6 = 15$

6. (B)

According to Stiffen law of cooling

$$\frac{d\theta}{dt} = k(\theta - \theta_0)$$

For case I,

$\theta = \frac{60° + 40°}{2}$ and $\theta_0 = 25°$

$$\frac{60° - 40}{15 - 0} = k\left[\frac{60 + 40}{2} - 25\right] \quad ....(1)$$

For case II,

$$\frac{40 - x}{30 - 15} = k\left[\frac{40 + x}{2} - 25\right] \quad ....(2)$$

Divide equation (1) by (2), we get

$$\frac{20}{15} \times \frac{15}{40 - x} = \frac{k(50 - 25)}{k\left(\frac{40 + x}{2} - 25\right)}$$

$\Rightarrow$ $25(40 - x) = 20\left[\frac{40 + x}{2} - 25\right]$

$\Rightarrow$ $1000 - 25x = 400 + 10x - 500$

$\Rightarrow$ $35x = 1100$

$\Rightarrow$ $x = \frac{1100}{35} = 31.43°C$

7. (A)

$f(x) = x^3 + 4x - 9 = 0$

According to Newton's-Raphson's method,

$$x_{k+1} = x_k - \frac{f(x)}{f'(x)}$$

$$\Rightarrow \quad x_{k+1} = x_k - \frac{x_k^3 + 4x_k - 9}{3x_k^2 + 4}$$

$$\Rightarrow \quad x_{k+1} = \frac{3x_k^3 + 4x_k - x_k^3 - 4x_k + 9}{3x_k^2 + 4}$$

$$\Rightarrow \quad 2x_{k+1} = \frac{2k_1^3 + 9}{3k_k^2 + 4}$$

8. (B)

$$L\left(\frac{\sin at}{t}\right) = \cot^{-1}\left(\frac{s}{a}\right)$$

$$\int_0^\infty e^{-st}\frac{\sin(at)}{t}dt = \cot^{-1}\left(\frac{s}{a}\right)$$

Put a = 1 and s = 0

$$\therefore \quad \int_0^\infty \frac{\sin t}{t}dt = \cot^{-1}(0) = \frac{\pi}{2}$$

9. (A)

Potential function, $\phi = x^2 - y^2$

$$\therefore \quad \frac{\partial\phi}{\partial x} = 2x = U$$

and $\dfrac{\partial\Psi}{\partial y} = U = 2x$

$$\frac{\partial\Psi}{\partial y} = 2x$$

Integrating $\int \partial\Psi = \int 2x\partial y$

$\therefore \quad \Psi = 2xy + c$

where, Ψ = stream function

At, $x = y = 0, \quad \Psi = 0$

$\Rightarrow \quad c = 0$

$\therefore \quad \Psi = 2xy$

10. (A)

Given Matrix, $A = \begin{bmatrix} 1 & 2 \\ 5 & 7 \end{bmatrix}$,

co-factors are $a_{11} = 7$, $a_{21} = -2$, $a_{12} = -5$, $a_{22} = 1$

$\therefore$ Matrix by co-factors = $\begin{bmatrix} 7 & -5 \\ -2 & 1 \end{bmatrix}$

$\therefore$ Adjoint A = $\begin{bmatrix} 7 & -2 \\ -5 & 1 \end{bmatrix}$

Value of determinant of matrix A, |A|

$= 1 \times 7 - 2 \times 5 = -3$

$$A^{-1} = \frac{\text{Adjoint A}}{|A|} = \frac{1}{-3}\begin{bmatrix} 7 & -2 \\ -5 & 1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} -7 & 2 \\ 5 & -1 \end{bmatrix}$$

11. (A)

Given, $x^3 - 10x^2 + 31x - 30 = 0$

Its one root is 5, hence

$(x - 5)(x^2 - 5x + 6) = 0$

$\Rightarrow (x - 5)(x^2 - 3x - 2x + 6) = 0$

$\Rightarrow (x - 5)(x - 3)(x - 2) = 0$

$\Rightarrow x = 5, 3, 2$

12. (C)

Standard deviation of spot speed of vehicles = 8.8 kmph

Mean speed of vehicles = 33 kmph

Coefficient of variation of speed

$$= \frac{\text{standard deviation of spot speed}}{\text{Mean speed of vehicles}} = \frac{8.8}{33} = 0.2666$$

2008

1. (B)

$(PQ)^{-1} P$

$\Rightarrow Q^{-1} P^{-1} P$ …. since, $(PQ)^{-1} = Q^{-1} P^{-1}$

$\Rightarrow Q^{-1}$ …. since $P^{-1} P = I$ (Unit Matrix)

2. (A)

$\frac{d^2y}{dx^2} + y = 0$ or $(D^2 + 1)y = 0$ … $\left(D = \frac{d}{dx}\right)$

Auxiliary equation is,

$D^2 + 1 = 0$

$\therefore D = \pm i$

So its solution is, $y = C_1 e^{ix} + C_2 e^{-ix}$

Using Euler's theorem,

$y = C_1 e^0 (\cos x + i \sin x) + C_2 e^0 (\cos x - i \sin x)$

$y = (C_1 + C_2) \cos x + i (C_1 - C_2) \sin x$ … (i)

Now let $C_1 + C_2 = P$ and $i(C_1 - C_2) = Q$ … (ii)

From equations (i) and (ii), we get

$y = P \cos x + Q \sin x$

3. (D)

Substituting $x_t = x\sqrt{\frac{k_z}{k_x}}$

we get $\frac{\partial^2 h}{\partial x t^2} + \frac{\partial^2 h}{\partial z^2} = 0$

4. (A)

$$\int_0^3\int_0^x (6-x-y)\,dx\,dy = \int_0^3\left(6y - xy - \frac{y^2}{2}\right)_0^x dx = \int_0^3\left(6x - x^2 - \frac{x^2}{2}\right)dx = \left(3x^2 - \frac{x^3}{3} - \frac{x^3}{6}\right)_0^6$$

$= 13.5$

5. (D)

Two equations to evaluate a and b in equation
y = a + bx are:
$\sum y = Na + b\sum_x \Rightarrow 21 = 3a + 6b$
$\sum xy = a\sum_x + b\sum x^2 \Rightarrow 23 = 3a + 7b$
Solving we get,
a = 3, b = 2

6. (D)

$$\frac{dy}{dx} = \frac{-x}{y}$$

$\Rightarrow \quad y \cdot dy = -x\,dx$

Integrating, $\frac{y^2}{2} = \frac{-x^2}{2} + c$

But at x = 1,y = $\sqrt{3}$. therefore c = 2

So the solution is, $\frac{y^2}{2} = \frac{-x^2}{2} + 2$

or $\quad y^2 + x^2 = 4$

7. (B)

$$P\left(-\frac{1}{3} \le x \le \frac{1}{3}\right) = \int_{-1/3}^{1/3} x^2 dx = \left[\frac{x^3}{3}\right]_{-1/3}^{1/3} = 0.0247 = 2.47\%$$

8. (B)

Given $\quad [P] = \begin{bmatrix} 4 & 5 \\ 2 & -5 \end{bmatrix}$

then to find the eigen values

$$\begin{vmatrix} 4-\lambda & 5 \\ 2 & -5-\lambda \end{vmatrix} = 0$$

$\Rightarrow \quad (4 - \lambda)(-5 - \lambda) - 10 = 0$
$\Rightarrow \quad \lambda^2 + \lambda - 30 = 0$
Solving we get $\quad \lambda = -6, 5$ (eigen values)

9. (A)

10. (D)

The system will not have a unique solution if $\Delta = 0$

i.e. $\quad \Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & k \end{vmatrix} = 0$

or $\quad 1(2k - 12) - 1(k - 3) + 1(4 - 2) = 0$
or $\quad k = 7$

11. (C)
When two vectors are perpendicular to each other, their dot multiplication is zero.
i.e. $\vec{P}\cdot\vec{Q}=0$ and $\vec{P}$ and $\vec{Q}$ are perpendicular.

2009

1.	(A)	2.	(B)	3.	(D)
4.	(B)	5.	(C)	6.	(A)
7.	(A)	8.	(A)	9.	(B)

2010

1. (A)

$$\lim_{x\to 0}\frac{\sin(2/3^x)}{x}$$

0|0 form, so apply L–Hospital's rule

$$\lim_{x\to 0}\frac{\frac{2}{3}\cos(2/3^x)}{1}$$

Taking limit, we get

$$\frac{2}{3}\cos 0^\circ = \frac{2}{3}$$

2. (C)

$$\text{Probability} = \frac{\text{Number of outcomes favourable}}{\text{Total number of possible outcomes}}$$

Total number of possible outcomes
(HH, TT, HT, TH) = 4
Number of outcomes favourable (HH) = 1

$\therefore$ Required probability = $\frac{1}{4}$

3. (A)

Given: $\frac{d^3y}{dx^3}+4\sqrt{\left(\frac{dy}{dx}\right)^3+y^2}=0$

$$\Rightarrow \quad \frac{d^3y}{dx^3}=-4\sqrt{\left(\frac{dy}{dx}\right)^3+y^2}$$

$$\Rightarrow \quad \left(\frac{d^3y}{dx^3}\right)^2=16\left[\left(\frac{dy}{dx}\right)^3+y^2\right]$$

$$\Rightarrow \quad \left(\frac{d^3y}{dx^3}\right)^2-16\left(\frac{dy}{dx}\right)^3-16y^2=0$$

So, order = 3 and degree = 2

4. (C)

Given: $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0$

$\therefore \quad (D^2 + D - 6)\, y = 0$

It's auxiliary equation is

$\Rightarrow \quad D^2 + D - 6 = 0$

$\Rightarrow \quad (D + 3)(D - 2) = 0$

$\Rightarrow \quad D = -3, 2$

Hence solution is, $y = C_1 e^{-3x} + C_2 e^{2x}$

5. (B)

Given: $A = \begin{bmatrix} 3+2i & i \\ -i & 3-2i \end{bmatrix}$

Now $A^{-1} = \frac{[\text{adj} \cdot A]^T}{|A|}$

where $[\text{adj } A] = \begin{bmatrix} 3-2i & +i \\ -i & 3+2i \end{bmatrix}$

$\Rightarrow \quad [\text{adj } A]^T = \begin{bmatrix} 3-2i & -i \\ i & 3+2i \end{bmatrix}$

and $\quad |A| = (3 + 2i)(3 - 2i) - (i)(-i)$

$= 9 - 4i^2 + i^2$

$= 9 + 4 - 1 = 12$

$\therefore \quad A^{-1} = \frac{1}{12}\begin{bmatrix} 3-2i & -i \\ i & 3+2i \end{bmatrix}$

6. (A)

By Simpson's rule,

$$\int_h^b f(x)\,dx \approx \frac{h}{3}(f_0 + 4f_1 + 2f_2 + 4f_3 + \ldots + 2f_{2m-2} + 4f_{2m-1} + f_{2m})$$

Here, $h = 0.25$, $f_0 = 1$, $f_1 = 0.9412$, $f_2 = 0.8$, $f_3 = 0.64$, $f_4 = 0.50$

$$\therefore \quad \int_0^1 f(x)\,dx \approx \frac{0.25}{3}(1 + 4 \times 0.9412 + 2 \times 0.8 + 4 \times 0.64 + 0.5)$$

$= 0.7854$

Alternately

Given: $h = 0.25$.

Hence by Simpson's rule

$$\int_{x_0} f(x)\,dx = \frac{h}{3}[y_0 + y_n + 4(y_1 + y_3 + y_{n-1}) + 2(y_2 + y_4 + \ldots y_{n-2}]$$

$$= \frac{0.25}{3}[1 + 0.5 + 4(0.9412 + 0.64) + 2(0.8)]$$

$= 0.7854$

7. (C)

Given: $z = ax + by + ab$ …(i)

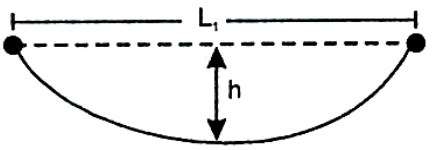
$p = \frac{\partial z}{\partial x} = a$ and $q = \frac{\partial z}{\partial y} = b$

Putting values of a and b in equation (i), we get

$z = px + qy + pq$

8. (D)

Given: $y = 4h\frac{x^2}{L^2}$



$\therefore \quad \frac{dy}{dx} = \frac{8hx}{L^2}$

Now, $\frac{ds}{dx} = \sqrt{1+\left(\frac{dy}{dx}\right)^2}$

$\therefore \quad s = 2\int_0^{L/2} 1 + \left(\frac{dy}{dx}\right)^2 dx$

$= 2\int_0^{L/2} \sqrt{\left(1+64\frac{h^2x^2}{L^4}\right)}\, dx$

9. (A)

Given: $f(x, y) = 4x^2 + 6y^2 - 8x - 4y + 8$

$\therefore \quad \frac{\partial f(x,y)}{\partial x} = 8x - 8$

Now $\frac{\partial f(x,y)}{\partial x} = 0$

$\Rightarrow \quad 8x - 8 = 0$

$\Rightarrow \quad x = 1$

and $\frac{\partial^2 f(x,y)}{\partial x^2} = 8$ (+ve)

So, at x = 1, f(x, y) is minimum.

$\therefore \quad \frac{\partial f(x,y)}{\partial y} = 12y - 4 = 0$

$\Rightarrow \quad y = \frac{1}{3}$

and $\frac{\partial^2 + (x,y)}{\partial y^2} = 12$ (+ve)

So, at $y = \frac{1}{3}$, f(x, y) is minimum

Putting the value of x = 1, $y = \frac{1}{3}$, we get

$$f(x, y) = 4 \times 1^2 + 6 \times \frac{1}{9} - 8 - 4 \times \frac{1}{3} + 8$$

$$= \frac{10}{3} \text{ (minimum)}$$

2011

1. (D)

$$[A] = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix}$$

$$[A]^T = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix}$$

$$[S] = \begin{bmatrix} 2a_{11} & a_{12} + a_{21} \\ a_{21} + a_{12} & 2a_{22} \end{bmatrix}$$

$$[D] = \begin{bmatrix} 0 & a_{12} - a_{21} \\ a_{21} - a_{12} & 0 \end{bmatrix}$$

For symmetric matrix, $[S] = [S]^T$, $[D] = [D]^T$
For skew-symmetric matrix, $[S] = -[S]^T$, $[D] = -[D]^T$
As it can be seen that both matrix [S] and [D] are symmetric only.

2. (A)

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

$$f(x) = x^2 - N$$

$$\Rightarrow \quad f(x_i) = 2x_i,$$

and $\quad f(x_i) = x_i^2 - N$

$$\therefore \quad x_{i+1} = x_i - \frac{x_i^2 - N}{2x_i} = \frac{1}{2}\left[x_i + \frac{N}{x_i}\right]$$

3. (C)

$$P(AB) = P(A) \cdot P(B) = \frac{4}{7} \times \frac{3}{7} = \frac{12}{49}$$

4. (D)

5. (C)

For a function f(x) to be continuous, at x = a

$$\Rightarrow \quad \underset{x \to \frac{\pi}{2}}{\text{Lt}} \frac{\lambda \cos x}{\frac{x}{2} - x} = 1$$

$$\Rightarrow \quad \underset{x \to \frac{\pi}{2}}{\text{Lt}} \frac{(-\lambda \sin x)}{-1} = 1$$

$\Rightarrow \quad \lambda \cdot \sin\frac{\pi}{2} = 1$

$\Rightarrow \quad \lambda = 1$

6. (B)

Given: $I = \int_0^a \frac{\sqrt{x}}{\sqrt{x}+\sqrt{a-x}} \cdot dx$ …(i)

Putting $a - x = t$, we get

$I = -\int_a^0 \frac{\sqrt{a-t}}{\sqrt{a-t}+\sqrt{t}} \cdot dt$

$= \int_0^a \frac{\sqrt{a-t} \cdot dt}{\sqrt{a-t}+\sqrt{t}}$

$= \int_0^a \frac{\sqrt{a-t}}{\sqrt{a-x}+\sqrt{x}} \cdot dx$ …(ii)

Adding equations (i) and (ii), we have

$2I = \int_0^a \frac{\sqrt{x}+\sqrt{a-x}}{\sqrt{x}+\sqrt{a-x}} \cdot dx$

$= \int_0^a dx = 0$

$\Rightarrow \quad I = \frac{a}{2}$

7. (A)

$\left|\bar{a}\times\bar{b}\right|^2 = (ab \sin\theta)^2 = (ab)^2 (1 - \cos^2\theta)$

$\Rightarrow \quad a^2b^2 - \left|\bar{a}\times\bar{b}\right|^2$

8. (D)

$\frac{dy}{dx} + \frac{y}{x} = x$

Putting $\nu = y^{-1}$

$\Rightarrow \quad d\nu = -y^{-2}dy = -\nu^2 \cdot dy$

$\therefore \quad \frac{-\nu(\nu^{-2}d\nu)}{dx} + \frac{y}{x} = x$

$\Rightarrow \quad -(\nu^{-1}x)\frac{d\nu}{dx} + y = x^2$

$\Rightarrow \quad -(\nu^{-1}x)\frac{d\nu}{dx} + \nu^{-1} = x^2$

$\Rightarrow \quad + \int d(\nu^{-1}x) = \int x^2 dx$

$\Rightarrow \quad v^{-1}x = \frac{x^3}{3} + c$

$\Rightarrow \quad y = \frac{x^2}{3} + \frac{c}{3x}$

At x = 1, y = 1

$\therefore \quad 1 = \frac{1}{3} + \frac{c}{3}$

$\Rightarrow \quad c = 2$

Hence $y = \frac{x^2}{3} + \frac{2}{3x}$

2012

1. (D)

Exact value of $\int_{0.5}^{1.5} \frac{1}{x}\, dx = 1.0986$

Using Simpson's rule in three-point form,

$h = \frac{b-a}{2} = \frac{1.5-0.5}{2} = 0.5$

So,

x	0.5	1	1.5
y	2	1	0.67

$\int \frac{1}{x}\, dx = \frac{h}{3}\,[y_0 + y_n + 4_{y1}]$

$= \frac{0.5}{3}\,[2 + 0.67 + 4 \times 3] = 1.1116$

It exceeds exacts by 0.012 (approx.)

2. (A)

$1 - P(7) = 1 - \left(\frac{1200-1000}{200}\right)$

$= 1 - P\left(\frac{1}{T}\right) = 0.159 < 50\%$

3. (B)

e^x

Put x = 2 in given equation which comes to about 7 and $e^2 = 7$

4. (D)

5. (D)

$p = q = \frac{1}{2}$

probability = $n_{C_r}\ p^r q^{n-r}$

$= {}^5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^5 + {}^5C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^4 = \frac{6}{32}$

6. (B)

$$\begin{vmatrix} 9-\lambda & 5 \\ 5 & 8-\lambda \end{vmatrix} = 0$$

$\Rightarrow (9 - \lambda)(8 - \lambda) = 25$

$\Rightarrow \lambda = 3.48, 13.53$

7. (A)

Area of parallelogram,

$A = \overline{OP} \times \overline{OR} = k(ad - bc)$

$$\overline{OP} \times \overline{OR} = \begin{vmatrix} i & j & k \\ a & b & 0 \\ c & d & 0 \end{vmatrix}$$

$|A| = ad - bc$

8. (D)

Given: $\frac{dy}{dx} + 2y = 0$

and at x = 1, y = 5

Differential equation is,

$(D + 2)y = 0$

$\Rightarrow \quad D = -2$

It's solution is $y = C_1 e^{-2x}$

At x = 1,

$5 = c_1 e^{-2}$

$\Rightarrow \quad 5e^2 = c_1 = 36.95$

Hence required solution is

$y = 36.95e^{-2x}$

2013

1.	0.5	2.	16	3.	0.52 to 0.55
4.	(B)	5.	6		

2014

1. (C)

Given, $\lim_{x\to\infty}\left(\frac{x+\sin x}{x}\right)$

Put $x = \frac{1}{h}$ and $x \to \infty$ or $h \to 0$

Now, $\lim_{h\to 0}\left(\frac{\frac{1}{h}+\sin\frac{1}{h}}{\frac{1}{h}}\right) = \lim_{h\to 0}\left(\frac{1+h\sin\frac{1}{h}}{1}\right)$

$= \frac{1+0}{1} = 1$

2. 23 to 23

Given, Matrices $J = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 6 \end{bmatrix}$ and $K = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$

$$K^T = \begin{bmatrix} 1 & 2 & -1 \end{bmatrix}$$

$$\therefore \quad K^TJK = \begin{bmatrix} 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 3+4-1 & 2+8-2 & 1+4-6 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 8 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = 6 + 16 + 1$$

$$K^TJK = 23$$

3. 0.4 to 0.4

Given, probability density function of evaporation E is

$$f(E) = \begin{cases} \frac{1}{5}, & 0 \le E \le 5 \text{ mm / day} \\ 0, & \text{otherwise} \end{cases}$$

As, the probability has to be find between 2 and 4 mm/day, therefore

$$P(2 < E < 4) = \int_2^4 f(E)\,dE = \int_2^4 \frac{1}{5}\,dE = \frac{1}{5}[E]_2^4 = \frac{4-2}{5} = \frac{2}{5}$$

$\therefore \quad P(2 < E < 4) = 0.4$

4. (A)

Given, Matrix $[M] = \begin{bmatrix} 215 & 650 & 795 \\ 655 & 150 & 835 \\ 485 & 355 & 550 \end{bmatrix}$

Sum of the eigenvalues = Trace of matrix

= 215 + 150 + 550 = 915

5. (A)

6. 0.26 to 0.27

Given, mean $\lambda = 5$

The probability that there will be less than 4 penalties in a day is

$P(y < 4) = P(y = 0) + P(y = 1) + P(y = 2) + P(y = 3)$

$$= \frac{e^{-5}5^0}{0!} + \frac{e^{-5}5^1}{1!} + \frac{e^{-5}5^2}{2!} + \frac{e^{-5}5^3}{3!} = e^{-5}\left[\frac{5^0}{0!} + \frac{5^1}{1!} + \frac{5^2}{2!} + \frac{5^3}{3!}\right]$$

= 0.006737 [1 + 5 + 12.5 + 20.833]

= 0.006737 × 39.33 = 0.2649

7. (B)

The probability of getting 'tail' when the coin is tossed again is $\frac{1}{2}$

8. 88 to 88

$$A = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 3 & 0 \\ 2 & 3 & 0 & 1 \\ 3 & 0 & 1 & 2 \end{bmatrix}$$

$$\Delta = -1\begin{vmatrix} 1 & 3 & 0 \\ 2 & 0 & 1 \\ 3 & 1 & 2 \end{vmatrix} + 2\begin{vmatrix} 1 & 0 & 0 \\ 2 & 3 & 1 \\ 3 & 0 & 2 \end{vmatrix} - 3\begin{vmatrix} 1 & 0 & 3 \\ 2 & 3 & 0 \\ 3 & 0 & 1 \end{vmatrix}$$

$= -1[1(-1) - 3(4 - 3)] + 2[1(6)] - 3[1(3) + 3(-9)]$

$= -1[-1 - 3] + 12 - 3[3 - 27]$

$= 4 + 12 - 3[-24] = 4 + 12 + 72 = 88.$

9. (B)

$$Z = \frac{2-3i}{-5+i}$$

Now by conjugating

$$Z = \frac{2-3i}{-5+i} \times \frac{(-5-i)}{(-5-i)} = \frac{-10+13i+3i^2}{25-i^2}$$

$$Z = \frac{-10+13i-3}{25-(-1)} = \frac{-13+13i}{26} = -0.5 + 0.5\,i$$

10. (D)

The integrating factor for the differential equation $\frac{dp}{dt} + K_2 p = K_1 L_0 e^{-K_1 t}$ is $e^{K_2 t}$

11. (B)

12. (A)

$$\lim_{\alpha \to 0} \frac{x^{\alpha} - 1}{\alpha} \left[\frac{0}{0} \text{ form}\right]$$

Hence applying L Hospital rule

Let $y = x^{\alpha}$

$\therefore$ $\log y = \alpha \log x$

$$\frac{1}{y}\frac{dy}{dx} = \frac{\alpha}{x} + \log x \frac{d\alpha}{dx}$$

$$\therefore \quad \frac{dy}{dx} = x^{\alpha}\left[\frac{\alpha}{x} + \log x \frac{d\alpha}{dx}\right]$$

Now applying L Hospital rule

$$\lim_{\alpha \to 0} \frac{x^{\alpha}\left[\frac{\alpha}{x} + \log x \frac{d\alpha}{dx}\right]}{\frac{d\alpha}{dx}} = \log x$$

13. 0.25 to 0.28

14. 2.0 to 2.0

The given matrix is

$$A = \begin{bmatrix} 6 & 0 & 4 & 4 \\ -2 & 14 & 8 & 18 \\ 14 & -14 & 0 & -10 \end{bmatrix}$$

The matrix is of order 3×4

Hence $\rho(A) \le 3$

$$\begin{vmatrix} 6 & 0 & 4 \\ -2 & 14 & 8 \\ 14 & -14 & 0 \end{vmatrix} = 6(0 + 112) + 4(28 - 196)$$

$$= 672 - 672 = 0$$

and $$\begin{vmatrix} 0 & 4 & 4 \\ 14 & 8 & 18 \\ -14 & 0 & -10 \end{vmatrix} = -4(-140 + 252) + 4(112)$$

$$= -4(112) + 4\,(112) = 0$$

Hence $\rho(A) \le 2$.

Now $\begin{vmatrix} 6 & 0 \\ -2 & 14 \end{vmatrix} = 84 \ne 0$

Hence $\rho(A) = 2$

$\therefore$ Rank of given matrix is 2

2015

1. 4.49 to 4.51

The given equation are

$2x + 3y = 5$ and $3x + Py = 10$

$$\therefore \quad \begin{bmatrix} 2 & 3 \\ 3 & P \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

Now comparing with AX = B

Augmented matrix $[A/B] = \begin{bmatrix} 2 & 3 & 5 \\ 3 & P & 10 \end{bmatrix}$

Now applying $R_2 \to 2R_2 - 3R_1$,

We get $\begin{bmatrix} 2 & 3 & 5 \\ 0 & 2p-9 & 5 \end{bmatrix}$

If $\int(A/B) \ne \int(A)$, then system will have no solution.

$\therefore \quad 2P - 9 = 0$

$\therefore \quad P = \dfrac{9}{2} = 4.5$

2. (A)

By trapezoidal rule,

Approximated value of $\int_a^b f(x)\,dx$ is always greater than analytical value

$$J > I$$

Here J is approximate value and I is analytical value.

3. 0.23 to 0.25

According to question

x	0	1
P (x, q)	q	1 – q

Now q = 0.4 (given)

∴

x	0	1
P (x, q)	0.4	(1 – 0.4) = 0.6

Now variance of x i.e. $v(x) = E(x^2) - [E(x^2)]^2$

Now $E(x) = \Sigma x_i P(x, q) = 0 \times 0.4 + 1 \times 0.6$

$= 0 + 0.6 = 0.6$

and $E(x^2) = \Sigma x_i^2 P(x, q)$

$= (0)^2 \times 0.4 + (1)^2 \times 0.6$

$= 0 + 0.6 = 0.6$

∴ $V(x) = E(x) - [E(x^2)]^2$

$= 0.6 - (0.6)^2$

$= 0.6 - 0.36 = 0.24$

4. (D)

The given matrix is

$$A = \begin{bmatrix} 3 & -2 & 2 \\ 4 & -4 & 6 \\ 2 & -3 & 5 \end{bmatrix}$$

In order to find Eigen value of above matrix

$$|A - \lambda I| = 0$$

$$\therefore \quad \begin{bmatrix} 3-\lambda & -2 & 2 \\ 4 & -4-\lambda & 6 \\ 2 & -3 & 5-\lambda \end{bmatrix} = 0$$

∴ $(3 - \lambda)[(-4 - \lambda)(5 - \lambda) + 18] + 2[20 - 4\lambda - 12] + 2[-12 + 8 + 2\lambda] = 0$

$(3 - \lambda)[-20 + 4\lambda - 5\lambda + \lambda^2 + 18] + 2[8 - 4\lambda] + 2[-4 + 2\lambda] = 0$

∴ $(3 - \lambda)(\lambda^2 - \lambda - 2) + 16 - 8\lambda - 8 + 4\lambda = 0$

$3\lambda^2 - 3\lambda - 6 - \lambda^3 + \lambda^2 + 2\lambda + 16 - 8\lambda - 8 + 4\lambda = 0$

∴ $-\lambda^3 + 4\lambda^2 - 5\lambda + 2 = 0$

∴ $\lambda^3 - 4\lambda^2 + 5\lambda - 2 = 0$

$\lambda_1 = 2$ and $\lambda_2 = 1$

Thus smallest eigen value of matrix is 1 and largest eigen value of matrix is 2.

∴ Option (D) is correct.

5. 2.32 to 2.34

6. (C)

The differential equation

$$x(ydx+xdy)\cos\frac{y}{x}=y(xdy-ydx)\sin\frac{y}{x}$$

$$\therefore\quad x(ydx+xdy)\cos\frac{y}{x}+\left(\sin\frac{y}{x}\right)y(xdy-ydx)=0$$

$$\therefore\quad (ydx+xdy)\cos\left(\frac{y}{x}\right)+\left(-\sin\frac{y}{x}\right)y\left(\frac{xdy-ydx}{x^2}\right)=0$$

$$\therefore\quad (ydx+xdy)\cos\left(\frac{y}{x}\right)+(xy)\left(-\sin\frac{y}{x}\right)\left(\frac{xdy-ydx}{x^2}\right)=0$$

Now by integrating we get

$$xy\cos\left(\frac{y}{x}\right)=c$$

7. (A)

Here $f(z)=\dfrac{9}{(z-1)(z+2)^2}$

At z = 1, pole is of simple order and at z = –2, pole is of second order.

Residue at z = 1

$$=\lim_{z\to1}(z-1)\frac{9}{(z-1)(z+2)^2}$$

$$=\lim_{z\to1}\frac{9}{(z+2)^2}\frac{9}{(1+2)^2}=1$$

Residue at z = –2

$$=\lim_{z\to-2}\frac{d}{dz}\left[(z+2)^2\frac{9}{(z-1)(z+2)2}\right]$$

$$=\lim_{z\to-2}\frac{-9}{(z-1)^2}=\frac{-9}{(-2-1)^2}=\frac{-9}{9}=-1$$

Thus one of the residue of above function is –1.

8. –5.72 to –5.70

Here $u(x, y, z) = x^2 - 3yz$

Now directional derivative of u

$$\therefore\quad \nabla u=\left(\frac{\partial}{\partial x}\hat{i}+\frac{\partial}{\partial y}\hat{j}+\frac{\partial}{\partial z}\hat{k}\right)u$$

$$=\left(\frac{\partial}{\partial x}\hat{i}+\frac{\partial}{\partial y}\hat{j}+\frac{\partial}{\partial z}\hat{k}\right)(x^2-3yz)$$

$$=\frac{\partial}{\partial x}(x^2-3yz)\hat{i}+\frac{\partial}{\partial y}(x^2-3yz)\hat{j}+\frac{\partial}{\partial z}(x^2-3yz)\hat{k}$$

$$=2x\hat{i}-3z\hat{j}-3y\hat{k}$$

Now directional derivative of u in direction of vector $\hat{i}+\hat{j}-2\hat{k}$

i.e. $$\nabla u.\hat{a} = \frac{\left(2x\hat{i}-3z\hat{j}-3y\hat{k}\right).\left(\hat{i}+\hat{j}-2\hat{k}\right)}{\sqrt{1+1+4}}$$

$$= \frac{2x-3z+6y}{\sqrt{6}}$$

Now $$\nabla u.\hat{a}\Big|_{(2,-1,4)} = \frac{2(2)-3(4)+6(-1)}{\sqrt{6}}$$

$$= \frac{4-12-6}{\sqrt{6}}$$

$$= \frac{-14}{\sqrt{6}} = -\frac{7\sqrt{6}}{3} = -5.72$$

9.	(D)	10.	0.3 to 0.4	11.	(C)
12.	(D)	13.	(B)	14.	1.06 to 1.07
15.	18	16.	(D)	17.	1.36 to 1.37

2016

1. 0.355 to 0.365

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 0.333 - \frac{f(0.333)}{f'(0.333)}$$

f (0.333) = 3 (0.333) – $e^{0.333}$ + sin (0.333)
= 0.999 – 1.3951 + 0.3268
= – 0.0693

f′(0.333)= 3 – $e^{0.333}$ + sin (0.333)
= 3 – 1.3951 + 0.3268
= 2.5499

$$\therefore \quad x_1 = 0.333 + \frac{0.0693}{2.5499}$$

= 0.333 + 0.02717
= 0.3601

2. (C)

B^2 – 4ac = 9 – 4
= 5 > 0
∴ PDE is Hyperbolic

3. (D)

All options are correct.

4. (A)
Type II error means acceptance of the null hypothesis when it is false and should be rejected.

5. (B)

6. (A)

$$f(x) = \begin{cases} 0.25 & \text{if } 1 \le x \le 5 \\ 0 & \text{otherwise} \end{cases}$$

$$P(x \le 4) = \int_{-\infty}^{4} f(x)d(x) = \int_{-\infty}^{1} f(x)d(x) + \int_{1}^{4} f(x)d(x)$$

$$= \int_{1}^{4} \frac{1}{4}dx = \frac{1}{4}(x)_1^4 = \frac{3}{4}$$

7. (B)

$$\int_0^\infty \frac{1}{1+x^2}dx + \int_0^\infty \frac{\sin x}{x}dx$$

$$= \left[\tan^{-1}(x)\right]_0^\infty + \frac{\pi}{2}$$

$$= \frac{\pi}{2} + \frac{\pi}{2} = \pi$$

8. (B)
$y = x^2 + 1$; $x + y = 3$ area bounded by parabola
$x + y = 3$
$\Rightarrow$ $x + x^2 + 1 = 3$
$\Rightarrow$ $x^2 + x - 2 = 0$
$\Rightarrow$ $(x + 2)(x - 1) = 0$
$\Rightarrow x = -2, 1$

$$A = \iint dydx$$

$$= \int_{-2}^{1}\int_{x^2+1}^{3-x} dydx$$

$$= \int_{-2}^{1} [y]_{x^2+1}^{3-x} dx$$

$$= \int_{-2}^{1} \left(3 - x - x^2 - 1\right)dx$$

$$= \int_{-2}^{1} \left(2 - x - x^2\right)dx$$

$$= \left[2x - \frac{x^2}{2} - \frac{x^3}{3}\right]_{-2}^{1}$$

$$= \left(2 - \frac{1}{2} - \frac{1}{3}\right) - \left(-4 - 2 + \frac{8}{3}\right)$$

$$= \left(\frac{12 - 3 - 2}{6}\right) - \left(\frac{-10}{3}\right)$$

$$= \frac{7}{6} + \frac{10}{3} = \frac{27}{6} = \frac{9}{2}$$

9. (A)

$$\frac{d^4y}{dx^4} + 3\frac{d^2y}{dx^2} = 108x^2$$

$A = m^4 + 3m^2 = 0 \qquad \Rightarrow m = 0, 0, \pm\sqrt{3}\,i$

C.F. $= c_1 + c_2x + c_3 \sin\sqrt{3x} + c_4 \cos\sqrt{3}\,x$

$$PI = \left(\frac{1}{D^4 + 3D^2}\right)108x^2$$

$$= \frac{1}{3D^2\left[1 + \frac{D^4}{3D^2}\right]}108x^2$$

$$= \frac{1}{3D^2}\left[1 + \frac{D^2}{3}\right]^{-1} 108x^2$$

$$= \frac{1}{3D^2}\left[1 - \frac{D^2}{3} + ...\right]108x^2$$

$$= \frac{1}{3D^2}\left[108x^2 - 72\frac{x^2}{2}\right] = 3x^4 - 12x^2$$

$$y(x) = c_1 + c_2x + c_3 \sin\sqrt{3x} + c_4 \cos\sqrt{3}x + 3x^4 - 12x^2$$

10. 54.49 to 54.51

Increasing order of spot speeds

32, 45, 49, 51, 53, 56, 60, 62, 66, 79

Median = middle values average = $\frac{53 + 56}{2} = 54.5\,\text{kmph}$

11. (D)

$f(x) = x^2 - 4x + 2$

$f'(x) = 2x - 4 = 0 \Rightarrow x = 2$

$f''(x) = 2 > 0$ minimum at $x = 2$

$\therefore$ min value is $f(2) = 4 - 8 + 2 = -2$

12. (C)

$$f(0)=\frac{\pi}{4}+\frac{2}{\pi}\left(\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+.....\right)$$

$$\frac{f\left(0^-\right)+f\left(0^+\right)}{2}=\frac{\pi}{4}+\frac{2}{\pi}\left(1+\frac{1}{3^2}+\frac{1}{5^2}+.....\right)$$

$$\frac{\pi}{2}-\frac{\pi}{4}=\frac{2}{\pi}\left(1+\frac{1}{3^2}+\frac{1}{5^2}+.....\right)$$

$$1+\frac{1}{3^2}+\frac{1}{5^2}+.....=\frac{\pi^2}{8}$$

$$\Rightarrow \quad \sum_{n=1}^{\infty}\frac{1}{(2n-1)^2}=\frac{\pi^2}{8}$$

13. (A)

14. (D)

$$\lim_{x\to 0}\frac{xy}{x^2+y^2}$$

It is in the form $\frac{0}{0}$

Put y = mx

As $y \to 0$, $mx \to 0$

$$\lim_{x\to 0}\frac{x^2m}{x^2\left(1+m^2\right)}=\frac{m}{1+m^2}$$

For different values of m we get different limits it is not unique therefore limit does not exist.

15. (B)

$$(AB)=\begin{pmatrix}1 & 2 & -3 & a\\ 2 & 3 & 3 & b\\ 5 & 9 & -6 & c\end{pmatrix}$$

$(R_2 - 2R_1)$; $(R_3 - 5R_1)$

$$\approx\begin{pmatrix}1 & 2 & -3 & a\\ 0 & -1 & 9 & b-2a\\ 0 & -1 & 9 & c-5a\end{pmatrix}$$

$(R_3 - R_2)$

$$\approx\begin{pmatrix}1 & 2 & -3 & a\\ 0 & -1 & 9 & b-2a\\ 0 & 0 & 0 & c-b-3a\end{pmatrix}$$

$(c - b - 3a) = 0$

$3a + b - c = 0$

16. (B)

$$E(f(x)) = \int_{-\infty}^{\infty} xf(x)dx$$

$$= \int_{-a}^{0}\left(\frac{x^2}{a}+x\right)dx + \int_{0}^{a}\left(\frac{-x^2}{a}+x\right)dx$$

$$= \left(\frac{x^3}{3a}+\frac{x^2}{2}\right)_{-a}^{0} + \left(\frac{-x^3}{3a}+\frac{x^2}{2}\right)_{0}^{a}$$

$$= -\left(\frac{-a^2}{3}+\frac{a^2}{2}\right)+\left(\frac{-a^2}{3}+\frac{a^2}{2}\right)$$

$$= 0$$

$$E(g(x)) = \int_{-a}^{0}\frac{-x^2}{a}dx + \int_{0}^{a}\frac{x^2}{a}dx$$

$$= \left(-\frac{x^3}{3a}\right)_{-a}^{0} + \left(\frac{x^3}{3a}\right)_{0}^{a}$$

$$= -\left(\frac{a^2}{3}\right)+\frac{a^2}{3} = 0$$

$$V(f(x)) = E(X^2) - (E(X))^2$$

$$E(X^2) = \int_{-a}^{0}\left(\frac{x^3}{a}+x^2\right)dx + \int_{0}^{a}\left(\frac{-x^3}{a}+x^2\right)dx$$

$$= \left(\frac{x^4}{4a}+\frac{x^3}{3}\right)_{-a}^{0} + \left(-\frac{x^4}{4a}+\frac{x^3}{3}\right)_{0}^{a}$$

$$= -\left(\frac{a^3}{4}-\frac{a^3}{3}\right)+\left(-\frac{a^3}{4}+\frac{a^3}{3}\right)$$

$$E(X^2) = \frac{a^3}{6}$$

$$V(f(x)) = \frac{a^3}{6}$$

$$V(g(x)) = E(X^2) - (E(X))^2$$

$$E(X^2) = \int_{-a}^{0} -\frac{x^3}{a}dx + \int_{0}^{a}\frac{x^3}{a}dx = \left(-\frac{x^4}{4a}\right)_{-a}^{0} + \left(\frac{x^4}{4a}\right)_{0}^{a}$$

$$= (0) - \left(-\frac{a^3}{4}\right)+\frac{a^3}{4} = \frac{2a^3}{4} = \frac{a^3}{2}$$

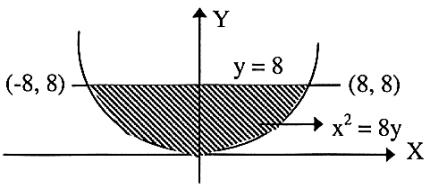
$$V(g(x)) = \frac{a^3}{2}$$

17. (D)
Angle between the curves is angle between the tangents at the point of intersection.

18. 85.0 to 85.5

Area = $2\int_0^8 \sqrt{8y}\,dy$

$$= 2\sqrt{8}\left.\frac{y^{\frac{3}{2}}}{\frac{3}{2}}\right]_0^8 = 85.33$$



19. (B)
Quadratic Approximation

$= f(0) + (x-0)f'(0) + \frac{(x-0)^2}{2!}f''(0)$

$= f(x) = x^3 - 3x^2 - 5 \Rightarrow f(0) = -5$

$f'(x) = 3x^2 - 6x \Rightarrow f'(0) = 0$

$f' = 6x - 6 \Rightarrow f''(x) = -6$

$\therefore$ Equation is $= -5 + x(0) + \frac{x^2}{2}(-6)$

$= -3x^2 - 5$

2017

1. (C)
Given P is inverse of Q
$\Rightarrow PQ = QP = I$

2. (B)
Probability density functions of univariable exponential distributes
$f(x) = \lambda e^{-dx}$ $\quad x \geq 0$
$= 0$ $\quad$ others
where λ is parameter

For Gaussian distribution $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{2}\right)^2}$

where μ and σ are parameters

3. (B)
$f(x) = e^{-x-e^{-x}}$ $\quad x \in (-\infty, \infty)$ is a continuous variable

$g(x) = \int f(x)\,dx = \int e^{-x-e^{-x}}dx = \int e^{-x}.e^{-e^{-x}}\,dx$

put $e^{-x} = t$
$-e^{-x}\,dx = dt$

$\therefore\ g(x) = \int e^{-t}(-dt) = -\left(\frac{e^{-t}}{-1}\right) = e^{-t} = e^{-e^{-x}}$

4. 35.9 to 36.1

Given $3\frac{\partial^2\phi}{\partial x^2}+B\frac{\partial^2\phi}{\partial x\partial y}+3\frac{\partial^2\phi}{\partial y^2}+4\phi=0$

By comparing with general form

$$A\frac{\partial^2\phi}{\partial x^2}+B\frac{\partial^2\phi}{\partial x\partial y}+C\frac{\partial^2\phi}{\partial y^2}+F\left(\phi,x,y,\frac{\partial u}{\partial x},\frac{\partial u}{\partial y}\right)=0$$

A = 3; B = B; C = 3

Condition for parabolic is

$B^2 - 4AC = 0$

$B^2 - 4(3 \times 3) = 0 \Rightarrow B^2 = 36$

$\therefore B^2 = 36$

5. –1.1 to –0.9

$$\lim_{x\to 0}\left(\frac{\tan x}{x^2-x}\right)=\lim_{x\to 0}\left(\frac{\tan x}{x(x-1)}\right)$$

$$=\lim_{x\to 0}\left(\frac{\tan x}{x}\right).\lim_{x\to 0}\left(\frac{1}{x-1}\right)=1\times(-1)=-1$$

6. (B)

f(x) = a + bx $0 \le x \le 1$ is a valid probability density function

i.e., $\int_0^1 f(x)\,dx=1$

$$\int_0^1(a+bx)\,dx=1$$

$$\left[ax+\frac{bx^2}{2}\right]_0^1=1$$

$$\Rightarrow a+\frac{b}{2}=1$$

$\Rightarrow 2a + b = 2$

a = 0.5, b = 1 satisfies the above function.

7. (D)

$\frac{d\theta}{dt}+\theta=1$ and $\theta = 0$ at t = 0

Comparing with first order linear differential equations

$\frac{dQ}{dt}+pQ=q$ where p = 1; q = 1

I.F = $\int_e pdt=e^t$

$$Q.(IF)=\int 1.(IF)\,dt+c$$

$$Q.e^t=\int e^t\,dt+c$$

$$Q.e^t=e^t+c$$

Q = 0 at t = 0 $\Rightarrow$ 0.1 = 1 + c $\Rightarrow$ c = –1

$\therefore\ Q.e^t=e^t-1\Rightarrow Q=1-e^{-t}$

8. (A)

Let A = $\begin{bmatrix} 5 & -1 \\ 4 & 1 \end{bmatrix}$

Characteristic equations is $\lambda^2 - 6\lambda + 9 = 0 \Rightarrow \lambda = 3, 3$

Eigen value 3 has multiplicity 2.

Eigen vectors corresponding to $\lambda = 3$ is $(A - 3I)\,X = 0$

$$\begin{pmatrix} 5-3 & -1 \\ 4 & 1-3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 2 & -1 \\ 4 & -2 \end{pmatrix}\begin{pmatrix} x \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$R_2 \to R_2 - 2R_1 \Rightarrow \begin{pmatrix} 2 & -1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

e(A) = 1

Number of linearly independent eigen vectors corresponding to eigen value $\lambda = 3$ is $n - r = 2 - 1 = 1$ where n = number of unknowns, r = rank of $(A - \lambda I)$

$\therefore$ One linearly independent eigen vector exists corresponding to $\lambda = 3$

9. 7.95 to 8.05

Approximation value by Euler's Method:

$\frac{du}{dt} = 3t^2 + 1$; u(0) = 0; h = Δt = 2

$u(2) = u(0) + hf(0, 0)$, $f(u, t) = 3t^2 + 1$

$= 0 + 2(0 + 1) = 2$

Exact value:

$du = (3t^2 + 1)dt$ (variable separable)

$\Rightarrow u = t^3 + t + c$ is solution

$u(0) = 0 \Rightarrow 0 = c$

$u = t^3 + t$

u(2) = 8 + 2 = 10

$\therefore$ absolute error = |10 – 2| = 8

10. (A)

Given systems

$3x_1 + 2x_2 = c_1$

$4x_1 + x_2 = c_2$

Matrix form is $\begin{pmatrix} 3 & 2 \\ 4 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$

AX = B

Characteristic equations of above systems is

$|A - \lambda I| = 0$

$$\begin{vmatrix} 3-\lambda & 2 \\ 4 & 1-\lambda \end{vmatrix} = 0$$

By expanding $\lambda^2 - 4\lambda - 5 = 0$

11. (C)

w = f(x, y) where x and y are functions of t.

By Chain rule,

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}.\frac{dx}{dt} + \frac{\partial w}{\partial y}.\frac{dy}{dt}$$

12. 133.9 to 134.1

Given

$v = x^2 i + 2y^3 J + z^4 k$

$$\text{div } v = \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(2y^3) + \frac{\partial}{\partial z}(z^4)$$

$= 2x + 6y^2 + 4z^3$

$$\text{div } v\big|_{(1,2,3)} = 2 + 24 + 108 = 134$$

13. 0.5 to 0.5

Given first three are already heads. If the coin is tossed again, the outcome does not depends on previous outcomes.

Probability getting head = $\frac{1}{2} = 0.5$

OR

Probability of first three are heads

= P(H × H × H) = $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$

Probability of fourth time head is

= P(H × H × H × H) = $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{16}$

Given condition is that (H.H.H) is already realized

The required probability = $\frac{1/16}{1/8} = 0.5$

14. (A)

$y = x\,\ell nx$

$$\frac{dy}{dx} = x.\frac{1}{x} + \ell nx$$

Given that $\theta = 45° \Rightarrow \tan\theta = 1$

$$\frac{dy}{dx} = \tan\theta = 1$$

i.e. $\Rightarrow x.\frac{1}{x} + \ell n\,x = 1 \Rightarrow 1 + \ell n\,x = 1$

x = 1, y = 0 satisfies

$\therefore$ A is correct.

15. (A)

Given

$$I = \int_0^1 \frac{(\sin^{-1} x)^2}{\sqrt{1-x^2}}\,dx$$

$$= \left.\frac{(\sin^{-1}x)^3}{3}\right|_0^1 \qquad \left(\because \int f^n(x)\, f^1(x)\, dx = \frac{f^{n+1}}{n+1}\right)$$

$$= \frac{1}{3}\left[(\sin^{-1})^3 - \sin^{-1}0\right] = \frac{1}{3}\left[\left(\frac{\pi}{2}\right)^3 - 0\right] = \frac{\pi^3}{24}$$

16. (A)

$$A = \begin{pmatrix} 1 & 5 \\ 6 & 2 \end{pmatrix} \qquad B = \begin{pmatrix} 3 & 7 \\ 8 & 4 \end{pmatrix}$$

$$AB^T = \begin{pmatrix} 1 & 5 \\ 6 & 2 \end{pmatrix}\begin{pmatrix} 3 & 8 \\ 7 & 4 \end{pmatrix} = \begin{bmatrix} 38 & 28 \\ 32 & 56 \end{bmatrix}$$

17. (A)

Given $y^{11} - 4y^1 + 3y = 2t - 3t^2$

$\Rightarrow (D^2 - 4D + 3)y = (2t - 3t^2)$

By definition of particular solution

$$y_p = \frac{1}{D^2 - 4D + 3}(2t - 3t^2)$$

$\Rightarrow (D^2 - 4D + 3)y_p = 2t - 3t^2$

verifying options, option (A) satisfies,

$(D^2 - 4D + 3)\ (-2 - 2t - t^2)$

$= -2 + 8 + 8t - 6 - 6t - 3t^2 = 2t - 3t^2$

$\therefore$ (A) is correct.

Alternate solution :

$y'' - 4y' + 3y = 2t - 3t^2$

$(D^2 - 4D + 3)y = 2t - 3t^2$

Particular solution = $y_p = \dfrac{1}{D^2 - 4D + 3}(2t - 3t^2)$

$$= \left[\frac{\frac{-1}{2}}{D-1} + \frac{\frac{1}{2}}{D-3}\right](2t - 3t^2)$$

$$= \frac{1}{2}\left[\frac{1}{1-D} - \frac{1}{3-D}\right]2\left(t - \frac{3}{2}t^2\right)$$

$$= \left[(1-D)^{-1} - \frac{1}{3}\left(1 - \frac{D}{3}\right)^{-1}\right]\left(t - \frac{3}{2}t^2\right)$$

$$= (1 + D + D^2 + D^3 \ldots)\left(t - \frac{3}{2}t^2\right)\frac{-1}{3}\left(1 + \frac{D}{3} + \frac{D^2}{9} + \ldots\right)\left(t - \frac{3}{2}t^2\right)$$

$$= \left[t - \frac{3}{2}t^2 + 1 - 3t - 3\right]\frac{-1}{3}\left[t - \frac{3}{2}t^2 + \frac{1 - 3t}{3} + \frac{1}{9}(-3)\right]$$

$$= -2 - 2t - \frac{-3}{2}t^2 - \frac{t}{3} + \frac{1}{2}t^2 - \frac{1}{9} + \frac{t}{3} + \frac{1}{9} = -2 - 2t - t^2$$

2018

1. (C)

Option (A): $|A| = 6 - 5 = 1$

Option (B): $|A| = 9 - 4 = 5$

Option (C): $|A| = 12 - 12 = 0$

Option (D): $|A| = 8 - 18 = -10$

Hence matrix (C) is singular

2. (C)

$$|Q| = \frac{3}{7}\left(-\frac{9}{49} - \frac{12}{49}\right) - \frac{2}{7}\left(\frac{18}{49} - \frac{4}{49}\right) + \frac{6}{7}\left(\frac{-36}{49} - \frac{6}{49}\right) = -1$$

$$\text{Adj. Q} = \begin{bmatrix} -\frac{21}{49} & \frac{42}{49} & -\frac{14}{49} \\ -\frac{14}{49} & -\frac{21}{49} & -\frac{42}{49} \\ -\frac{42}{49} & -\frac{14}{49} & \frac{21}{42} \end{bmatrix}$$

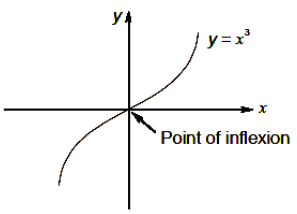
$$Q^{-1} = \frac{\text{AdjQ}}{|Q|} = \begin{bmatrix} \frac{3}{7} & -\frac{6}{7} & \frac{2}{7} \\ \frac{2}{7} & \frac{3}{7} & \frac{6}{7} \\ \frac{6}{7} & \frac{2}{7} & -\frac{3}{7} \end{bmatrix}$$

Or $\because$ Q is orthogonal

$\therefore Q^{-1} = Q^T$

3. (D)

$f(x) = x^3$ at $x = 0$



At $x = 0$, the function $y = x^3$ has neither minima nor maxima.

4. (B)

The value of $\int_0^{\pi} x\cos^2 x dx$

$$= \int_0^{\pi}\left(\frac{x}{2} + \frac{x\cos 2x}{2}\right)dx$$

$$= \frac{x^2}{4}\Big|_0^{\pi} + \frac{1}{2}\left(\frac{x\sin 2x}{2} + \frac{\cos 2x}{4}\right)$$

$$= \frac{\pi^2}{4} + \frac{1}{2}\left\{\left(0 + \frac{1}{4}\right) - \left(0 + \frac{1}{4}\right)\right\}$$

$$= \frac{\pi^2}{4} + \frac{1}{2}\left(\frac{1}{4} - \frac{1}{4}\right)$$

$$= \frac{\pi^2}{4}$$

5. (D)

$$\frac{c^2\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$$

u (x, 0) = f (x)

u_t (x, 0) = g (x)

$$\frac{\partial^2 u}{\partial x^2} = 25\frac{\partial^2 u}{\partial t^2}$$

f (x) = 3x

g (x) = 3

$c^2 = \frac{1}{25}$

D'Alembert's formula

$$u(x,t) = \frac{1}{2}\left[f(x+ct) + f(x-ct)\right] + \frac{1}{2c}\int_{x-ct}^{x+ct} g(y)\,dy$$

$$= \frac{1}{2}\left[3\left(x + \frac{t}{5}\right) + 3\left(x - \frac{t}{5}\right) + \frac{1}{2(1/5)}\int_{x-\frac{t}{5}}^{x+\frac{t}{5}} 3dy\right]$$

$$= \frac{1}{2}[6x] + \frac{3}{2}(5)\left[x + \frac{t}{5} - x + \frac{t}{5}\right]$$

u (x, t) = 3x + 3t

At x = 1, t = 1

u (x, t) = 6

6. 0.36 to 0.37

$(D^2 + 2D + 1)\, y = 0$ ($\therefore$ Roots are – 1, –1)

$CF = (C_1 + C_2 x)e^{-x}$

$y = C_1 e^{-x} + C_2 x e^{-x}$ …(i)

y(0) = 1 $\quad 1 = C_1$ …(ii)

$y = C_1 e^{-x} + C_2 (e^{-x} - xe^{-x})$

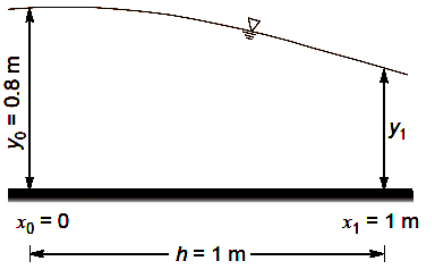
y'(0) = –1, $\quad -1 = -C_1 + C_2$ …(iii)

From eq. (ii) and (iii), $C_1 = 1$, $C_2 = 0$

$\therefore \quad y = e^{-x}$

At x = 1, $y = e^{-1} = \frac{1}{e} = 0.368$

7. 0.78 to 0.80



$$\frac{dy}{dx} = \frac{1 - e^{\frac{10}{3}\ln y}}{250 - 45e^{-3\ln y}}$$

$y_1 = y_0 + hf(x_0, y_0)$

$$\Rightarrow \quad y_1 = 0.8 + 1\left[\frac{1 - e^{-\frac{10}{3}\ln 0.8}}{250 - 45e^{-3\ln 0.8}}\right]$$

$$= 0.8 + 1\left(\frac{-1.1039}{162.109}\right) = 0.793 \text{ m}$$

8. (C)

$$x\frac{dy}{dx} + y = 0$$

$$x\frac{dy}{dx} = -y$$

$$\frac{dy}{y} = -\frac{dx}{x}$$

$$\int \frac{1}{y}dy = \int \frac{-1}{x}dx$$

$\ln y = -\ln x + c$

When, $y = 1, x = 1$

$c = 0$

$$\Rightarrow \quad y = \frac{1}{x} = x^{-1}$$

9. (A)

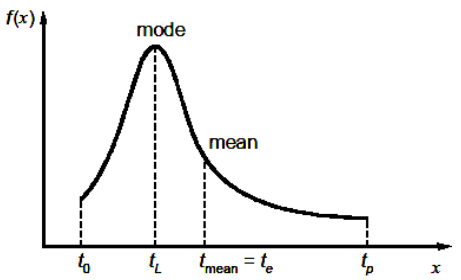
$$\int_0^3 f(x)dx = 1$$

$$\int_0^1 f(x)dx + \int_1^2 f(x)dx + \int_2^3 f(x)dx = 1$$

$$\frac{h}{2} + \frac{2h}{2} + \frac{3h}{2} = 1$$

$$6h = 2 \Rightarrow h = \frac{1}{3}$$

10. (C)



$t_L < t_{mean}$ = Curve is skew to right
mode < mean
i.e., mean > median and mode
Mean is greater than the mode and the median. This is common for a distribution that is skewed to the right [i.e., bunched up toward the left and a 'tail' stretching toward the right].

11. 0.5 to 0.5

Probability, $\bar{P} = \frac{5}{6} \times \frac{4}{5} \times \frac{3}{4} = \frac{1}{2} = 0.5$

12. 1 to 1

Given $f(x) = 2x^2 - 3x + 3,\ x_0 = 2$
$f(x) = 4x - 3$

By Newton–Rapshon

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 2 - \frac{2(2)^2 - 3(2) + 3}{4(2) - 3}$$

$$= 2 - \frac{5}{5} = 1$$

13. (D)

$$A = \begin{bmatrix} 2 & -4 \\ 4 & -2 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & -4 \\ 4 & -2-\lambda \end{vmatrix} = 0$$

$-4 - 2\lambda + 2\lambda + \lambda^2 + 16 = 0$

$\lambda^2 + 12 = 0$

$\lambda = \pm 2\sqrt{3}i$ (Complex eigen values)

(1) $\lambda = 2\sqrt{3}i$

Consider $(A - \lambda I)\ X = 0$

$$\begin{bmatrix} 2-2\sqrt{3}i & -4 \\ 4 & -2-2\sqrt{3}i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$2 - 2\sqrt{3}i x_1 = 4x_2$

$$\frac{x_1}{4} = \frac{x_2}{2 - 2\sqrt{3}i}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 - 2\sqrt{3}i \end{bmatrix}$$

(2) $\lambda = -2\sqrt{3}i$

Consider $(A - \lambda I)X = 0$

$$\begin{bmatrix} 2+2\sqrt{3}i & -4 \\ 4 & -2+2\sqrt{3}i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2+2\sqrt{3}ix_1=4x_2$$

$$\frac{x_1}{4}=\frac{x_2}{2+2\sqrt{3}i}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}=\begin{bmatrix} 4 \\ 2+2\sqrt{3}i \end{bmatrix}$$

$\therefore$ Complex Eigen values and complex Eigen vectors.

14. (B)

$$L\left(e^{at}\right)=\frac{1}{S-a}$$

$$L\left(e^{at}\right)=\int_0^\infty e^{-st}e^{at}dt=\int_0^\infty e^{-(s-a)t}dt$$

$$=\frac{e^{-(s-a)t}}{-(s-a)}\Bigg|_0^\infty=-\frac{1}{s-a}(0-1)=\frac{1}{s-a}$$

15. (B)

$$A=\begin{pmatrix} 1 & 1 & 0 & -2 \\ 2 & 0 & 2 & 2 \\ 4 & 1 & 3 & 1 \end{pmatrix}$$

$R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 4R_1$

$$\begin{pmatrix} 1 & 1 & 0 & -2 \\ 0 & -2 & 2 & 6 \\ 0 & -3 & 3 & 9 \end{pmatrix}$$

$$R_3 \to R_3 - \frac{3}{2}R_2$$

$$\begin{pmatrix} 1 & 1 & 0 & -2 \\ 0 & -2 & 2 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

No. of non zero rows = 2
Rank of A = 2

16. 0.60 to 0.70

$$\bar{F}=x^2\,\bar{i}+y^2\,\bar{j}$$

$$\int \bar{F}.d\bar{r}=\int\left(x^2\,\bar{i}+y^2\,\bar{j}\right).\left(dx\,\bar{i}+dy\,\bar{j}\right)$$

$$=\int x^2dx+y^2dy$$

(0, 0) to (1, 1) line is y = x

$$=\int x^2dx+x^2dx=\int_0^1 2x^2dx$$

$$=2\left(\frac{x^3}{3}\right)\Bigg|_0^1=\frac{2}{3}=0.666$$

2019

1. (A)

$$\lim_{x\to 0}\frac{\sin 4x}{\sin 2x}=\lim_{x\to 0}\frac{4x}{2x}\frac{\sin 4x}{4x}\times\frac{2x}{\sin 2x}=\frac{4}{2}=2$$

$$\lim_{x\to 0}\frac{\tan x}{x}=1$$

2. (A)

Taylor series of f(x + h) at x.

$$f(x+h)=f(x)+(x+h-x)\,f'(x)+\frac{(x+h-x)^2}{2!}f''(x)+.....$$

$$=f(x)+hf'(x)+\frac{h^2}{2!}f''(x)+\frac{h^3}{3!}f'''(x)+.....$$

3. (B)

Let $y = x^{1/x}$

$$\log y=\frac{\log x}{x}$$

$$y=e^{\frac{\log x}{x}}$$

y maximum (or) minimum when,

$$f(x)=\frac{\log x}{x}\text{ is maximum (or) minimum}$$

(1) $$f'(x)=\frac{x\left(\frac{1}{x}\right)-\log x}{x^2}=\frac{1-\log x}{x^2}$$

(2) $f'(x) = 0$

$\Rightarrow$ $1 - \log x = 0$

$\log x = 1$

$x = e$

(3) $$f''(x)=\frac{x^2\left(-\frac{1}{x}\right)-2x(1-\log x)}{x^4}=\frac{-3x+2x\log x}{x^4}$$

(4) $$f''(e)=-\frac{e}{e^4}<0(\max)$$

$\therefore$ $y=\sqrt[x]{x}$ is maximum at $x = e$

4. (A)

$$\frac{\partial v}{\partial t}=\beta\frac{\partial^2 v}{\partial x^2}\qquad\text{...(i)}$$

Approximate time derivative in equation (i) with forward difference,

$$\frac{\partial v}{\partial t}=\frac{v_i^{n+1}-v_i^n}{\Delta t}+0(\Delta t)\qquad\text{...(ii)}$$

Note that the right hand side only in value v at $x = x_i$

Use the central difference approximation to $\left(\frac{\partial^2 v}{\partial x^2}\right)_{x_i}$ and evaluate all the terms at time n.

$$\left.\frac{\partial^2 v}{\partial x^2}\right|_{x_i} = \frac{V_{i-1}^n - 2v_i^n + v_{i+1}^n}{\Delta x^2} + 0\left(\Delta x^2\right) \qquad \ldots\text{(iii)}$$

Substituting equation (ii) in the left hand side of equation (i), substitute the equation (iii) into the right hand side of equation (i), and collect the truncation error terms to get

$$\frac{v_i^{(n+1)} - v_i^{(n)}}{\Delta t} = \beta\left[\frac{v_{i+1}^{(n)} - 2v_i^{(n)} + v_{i-1}^{(n)}}{(\Delta x)^2}\right] + 0(\Delta t) + 0\left(\Delta x^2\right)$$

$$\therefore \qquad \frac{v_i^{(n+1)} - v_i^{(n)}}{\Delta t} = \beta\left[\frac{v_{i+1}^{(n)} - 2v_i^{(n)} + v_{i-1}^{(n)}}{(\Delta x)^2}\right]$$

5. (5.9 to 6.1)

$$[xD^2 - 2xD + 2]y = 0$$

Put $x = e^z$

$xD = D_1$

$x^2D^2 = D_1(D_1 - 1)$

$[D_1(D_1 - 1) - 2D_1 + 2]y = 0$

$\left[D_1^2 - 3D_1 + 2\right]y = 0$

AE: $m^2 - 3m + 2 = 0$

$m = 1, 2$

$y = C_1 e^z + C_2 e^{2z}$

$e^z = x$

$y = C_1x + C_2x^2$

$y(1) = 0$ $\qquad y(2) = 2$

$y(1) = C_1 + C_2 = 0$

$y(2) = 2C_1 + 4C_2 = 2$

$C_1 = -1, C_2 = 1$ $\qquad y = x^2 - x$

$y(3) = 6$

6. (D)

$$x = \begin{bmatrix} 4 \\ -2 \\ -6 \end{bmatrix}$$

Euclidean norm length = $\sqrt{16 + 4 + 36} = \sqrt{56}$

7. (A)

$$L(\sin h\,(at)) = \frac{a}{s^2 - a^2}$$

8. (D)

$$\lim_{x\to 0}\frac{x\sin x}{1-\cos x}$$

$$\lim_{x\to 0}\frac{\sin x}{\frac{1-\cos x}{x}}=\lim_{x\to 0}\frac{\frac{\sin x}{x}}{\frac{1-\cos x}{x^2}}=\frac{1}{1/2}=2$$

9. (D)

$$\text{curl}\,\bar{F}=\begin{vmatrix}\bar{i} & \bar{j} & \bar{k}\\ \partial/\partial x & \partial/\partial y & \partial/\partial z\\ 2x^2y & 5z^2 & -4yz\end{vmatrix}$$

$$=\bar{i}(-4z-10z)-\bar{j}(0-0)+\bar{k}(0-2x^2)$$

$$=-14zi-2x^2k$$

10. (B)

11. (B)

Probability density function of uniform distribution is

$$f(x)=\frac{1}{y-x}$$

12. (C)

$$\text{Volume of water}=\frac{1}{3}\pi h^2(3r-h)$$

$$=\frac{1}{3}\pi\times 6^2\times(3\times 13-6)=396\pi$$

13. (D)

$$\frac{dy}{dx}(x\ln x)=y$$

$$\frac{dy}{y}=\frac{dx}{x\ln x}$$

$$\int\frac{dy}{y}=\int\frac{1}{x\ln x}dx+\ln k$$

$$\ln x=t$$

$$\frac{1}{x}dx=dt$$

$$\ln y=\int\frac{dy}{t}+\ln k$$

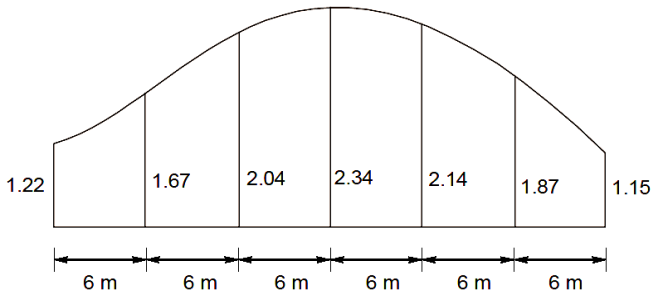
$$\ln y=\ln t+\ln k$$

$$\ln y=kt$$

$$y=k\ln x$$

14. (C)

15. (67.00 to 70.00)



Area by Simpson's rule

$$A = \frac{d}{3}\left[h_0 + h_h + 4(h_1 + h_3 + ...) + 2(h_2 + h_4 + ...)\right]$$

$$= \frac{6}{3}\left[1.22 + 1.15 + 4 \times (1.67 + 2.34 + 1.87) + 2(2.04 + 2.14)\right]$$

$$= 68.50 \text{ m}^2$$

2020

1. (C)

$\because$ 1st term of given D. Equation contains product of dependent variable with it's derivative, so it is non-linear and also we have 2nd order derivative so it's order is two i.e., 2nd order non linear equation.

2. (B)

$$\lim_{x \to 0} \frac{x^2 - 5x + 4}{4x^2 + 2x} = \frac{4}{0} = 0$$

Applying L's Hospital rule,

$$\lim_{x \to 0} \frac{f'(x)}{f'(x)} = \frac{2x - 5}{8x + 2} = -\text{ve}$$

So, again applying L's Hospital rule,

$$\lim_{x \to 0} f''(x) = \frac{2}{8} = \frac{1}{4}$$

3. (B)

TV = 0.69

$\ln_1 = 0$

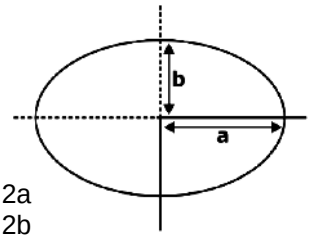
$\ln_6 = 1.79$

from interpolating ($\ln_1$ and $\ln_6$), we get = 0.358 (mv)

$$\% \text{ Error} = \frac{TV - mv}{TV} \times 100 = 48.11\%$$

4. (C)

Equation of ellipse is $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$



Major axis = 2a
Minor axis = 2b
Area of ellipse for standard equation is πab.

5. 0.60

T = 50 years

$P = \frac{1}{T} = \frac{1}{50}$

$q = 1 - P = \frac{49}{50}$

Reliability, $q^n = \left(\frac{49}{50}\right)^{25} = 0.6034 = 60.34\%$

6. (C)

$$\frac{d^2x}{dt^2}-\frac{5\,dx}{dt}+6x = 0$$

$D^2 - 5D + 6 = 0$

$D^2 - 3D - 2D + 6 = 0$

$D(D - 3) - 2(D - 3) = 0$

$D = 2, 3$

Solution will be, $x = C_1 e^{3t} + C_2 e^{2t}$

Applying boundary condition,

$x(0) = 0$

$C_1 + C_2 = 0$ …(1)

$\frac{dx}{dt} = 3C_1e^{3t} + 2C_2e^{2t}$

$\frac{dx(0)}{dt} = 10$

$10 = 2C_1 + 2C_2$ …(2)

From eqs. (1) and (2)

So, final solution becomes,

$x = 10e^{3t} - 10e^{2t}$

or $x = -10e^{2t} + 10e^{3t}$

7. (A)

$$\left.\frac{\partial^3 u}{\partial x^3}\right|_{x_i} = \frac{-u_{i+3}+8u_{i+2}-13u_{i+1}+13u_{i-1}-8u_{i-2}+u_{i-3}}{8\Delta h^3}$$

$$f'''(x_i) = \frac{-f(x_{i+3}) + 8f(x_{i+2}) - 13f(x_{i+1}) + 13f(x_{i-1}) - 8f(x_{i-2}) + f(x_{i-3})}{8h^3}$$

8. 3

Equation of a line segment,

$$\frac{x}{1} = \frac{y}{1} = \frac{z}{1} = k$$

$x = k, y = k, z = k \quad \Rightarrow \quad dx = dk$

$0 \le x \le 1$

$\therefore \quad 0 \le k \le 1$

$$\int_0^1 2k\ dk + \int_0^{1/2} k\ dk + \int_0^1 2k\ dk$$

$$= 3\left[k^2\right]_0^1$$

$$= 3$$

9. 3

Solving the matrix by multiplication,

$x_1 + 3x_2 + 2x_3 = 1$ …(1)

$2x_1 + 2x_2 - 3x_3 = 1$ …(2)

$4x_1 + 4x_2 - 6x_3 = 2$ …(3)

$2x_1 + 5x_2 + 2x_3 = 1$ …(4)

Solving eqs. (1) & (2) eliminating x_1, we get

$2x_1 + 6x_2 + 4x_3 = 2$

$2x_1 + 2x_2 - 3x_3 = 1$

$- \quad - \quad + \quad -$

$4x_2 + 7x_3 = 1$ …(5)

Solving eqs. (3) & (4) eliminating x_1, we get

$4x_1 + 4x_2 - 6x_3 = 2$

$4x_1 + 10x_2 + 4x_3 = 2$

$- \quad - \quad - \quad -$

$-6x_2 - 10x_3 = 0$ …(6)

Solving eqs. (5) & (6), we get

$4x_2 + 7x_3 = 1$

$-6x_2 - 10x_3 = 0$

$24x_2 + 42x_3 = 6$

$\underline{-24x_2 - 40x_3 = 0}$

$2x_3 = 6$

$x_3 = 3$

10. (B)

$$\frac{d^2u}{dx^2} - 2x^2u + \sin x = 0$$

The degree of all terms is different; hence it is non-homogeneous equation.
The given ordinary differential equation is linear because the coefficient of y and y′, y″ is the function of x.

11. (C)

$$\lim_{x\to\infty} \frac{\sqrt{9x^2 + 2020}}{x+7}$$

Dividing numerator and denominator by 'x'

$$\lim_{x\to\infty} \frac{\dfrac{\sqrt{9x^2+2020}}{x}}{\dfrac{x+7}{x}} = \lim_{x\to\infty} \frac{\sqrt{\dfrac{9x^2}{x^2} + \dfrac{2020}{x^2}}}{1+\dfrac{7}{x}} = \lim_{x\to\infty} \frac{\sqrt{9 + \dfrac{2020}{x^2}}}{1+\dfrac{7}{x}}$$

putting $x \to \infty \quad \Rightarrow \frac{1}{x^2} \to 0, \ \frac{1}{x} \to 0$

$$= \frac{\sqrt{9+0}}{1+0} = \frac{\sqrt{9}}{1} = 3$$

12. (D)

Simpson uses a cubic function for interpolation and its does not give an error for a cubic functions.

13. (C)

Let $u = xy$

$$\frac{\partial u}{\partial y} = xy' \qquad \frac{\partial^2 u}{\partial x^2} = yx'$$

Given : $\dfrac{\partial u}{\partial y} = \dfrac{\partial^2 u}{\partial x^2}$

$$xy' = yx''$$

$$\Rightarrow \quad \frac{y'}{y} = \frac{x''}{x} = k$$

$$y' - yk = 0 \qquad \ldots(1)$$

$$x'' - xk = 0 \qquad \ldots(2)$$

Take eq. (1)
Auxiliary equation is

$$D - k = 0 \qquad \Rightarrow \quad D = k$$

So the solution is

$$y = C_0 e^{ky}$$

Take eq. (2)

$$x'' - xk = 0$$

Auxiliary equation is

$D^2 - k = 0 \quad \Rightarrow \quad D = \pm\sqrt{k}$

So the solution is

$x = C_1 e^{\sqrt{k}x} + C_2 e^{-\sqrt{k}x}$

But $u = xy$

$= (C_1 e^{\sqrt{k}x} + C_2 e^{-\sqrt{k}x})\, C_0 e^{ky}$

$= C_0 C_1 e^{\sqrt{k}xky} + C_0 C_2 e^{-\sqrt{k}xky}$

$= k_1 e^{k^{3/2}xy} + k_2 e^{-k^{3/2}xy}$

Hence option (C).

14. 0.196

x = 15

$P(r = 8) = {}^nC_r\, p^r\, q^{n-r} \quad = {}^{15}C_8\, p^8\, q^{15-8}$

$= {}^{15}C_8\, p^8\, q^7$

p = (probability of success in toss) = $\frac{1}{2}$

q = (probability of failure during toss) = $\frac{1}{2}$

$P(r = 8) = {}^{15}C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^7 \quad = {}^{15}C_8 \left(\frac{1}{2}\right)^{15}$

$= 0.196$

15. (B)

$6\frac{d^2y}{dx^2} + \frac{dy}{dx} - y = 0$

$(6m^2 + m - 1)\, y = 0$

$m = \frac{d}{dx}$

Auxiliary equation is

$6m^2 + m - 1 = 0$

$6m^2 + 3m - 2m - 1 = 0$

$3m\,(2m + 1) - 1\,(2m + 1) = 0$

$(2m + 1)\,(3m - 1) = 0$

$m = -\frac{1}{2}, \frac{1}{3}$

$CF = C_1\, e^{-(1/2)x} + C_2\, e^{(1/3)x}$

$PI = 0 \quad$ as $f(x) = 0$

General solution,

$y = C_1\, e^{-(x/2)} + C_2\, e^{(x/3)}$

16. (D)

$$|P| = -(-2)\begin{vmatrix} 1 & 3 & 0 \\ 0 & 6 & 1 \\ 0 & 1 & 3 \end{vmatrix} \quad = 2\begin{vmatrix} 1 & 3 & 0 \\ 0 & 6 & 1 \\ 0 & 1 & 3 \end{vmatrix}$$

$= 2\{1(36-1) - 3(0-0) - 0(0-0)\}$
$= 70$

As, the determinant of the given matrix is non-zero.
Hence, the matrix is singular.
The product of eigen values = |A|

$\lambda_1, \lambda_2, \lambda_3, \lambda_4 = 70$

sum of eigen values = trace of matrix P

trace of matrix P $= 0 + 3 + 6 + 6$
$= 15$

$\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 50$

Checking the above options, only (D) option is correct.

$\lambda_1, \lambda_2, \lambda_3, \lambda_4 = 1, 2, 5, 7$

17. –6.58

Comparing the above equation with standard trigonometric Fourier's series representation,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$\frac{a_0}{2} = \frac{1}{T}\int_0^T f(x)\,dx, \quad -\pi < f(x) < \pi$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi}(x - x^2)\,dx \qquad = \frac{1}{2\pi}\left[\int_{-\pi}^{\pi} x\,dx - \int_{-\pi}^{\pi} x^2\,dx\right]$$

$$= \frac{1}{2\pi}\left[\left(\frac{x^2}{2}\right)_{-\pi}^{\pi} - \left(\frac{x^3}{3}\right)_{-\pi}^{\pi}\right] \qquad = \frac{1}{2\pi}\left[\left(\frac{\pi^2}{2} - \frac{\pi^2}{2}\right) - \left(\frac{\pi^3}{3} - \frac{-\pi^3}{3}\right)\right]$$

$$= \frac{1}{2\pi}\left[\frac{-2\pi^3}{3}\right] \quad = \frac{-1}{2\pi} \times \frac{-2\pi^3}{3} \quad = \frac{-\pi^2}{3}$$

$$a_0 = \frac{-2\pi^2}{3} \qquad = -6.58$$

2021

1. (B)

$$A = \begin{bmatrix} 1 & 2 & 2 & 3 \\ 3 & 4 & 2 & 5 \\ 5 & 6 & 2 & 7 \\ 7 & 8 & 2 & 9 \end{bmatrix}$$

Convert given matrix A into part row echelon form using elementary row transfer

$$\begin{matrix} R_2 \to R_2 - 3R_1 \\ R_3 \to R_3 - 5R_1 \\ R_4 \to R_4 - 7R_1 \end{matrix} \quad \Rightarrow \quad A = \begin{bmatrix} 1 & 2 & 2 & 3 \\ 0 & -2 & -4 & -4 \\ 0 & -4 & -8 & -8 \\ 0 & -6 & -12 & -12 \end{bmatrix}$$

$$\begin{matrix} R_4 \to R_4 - 7R_1 \\ R_4 \to R_4 - 3R_2 \end{matrix} \quad \Rightarrow \quad A = \begin{bmatrix} 1 & 2 & 2 & 3 \\ 0 & -2 & -4 & -4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Numbers of linearly independent non-zero rows in row-echelon form of A is 2 so rank at matrix A equal to 2.

2. (D)

We know that,

$$(PQ)^T = Q^T P^T$$

$$\Rightarrow \quad Q^T P^T = (PQ)^T$$

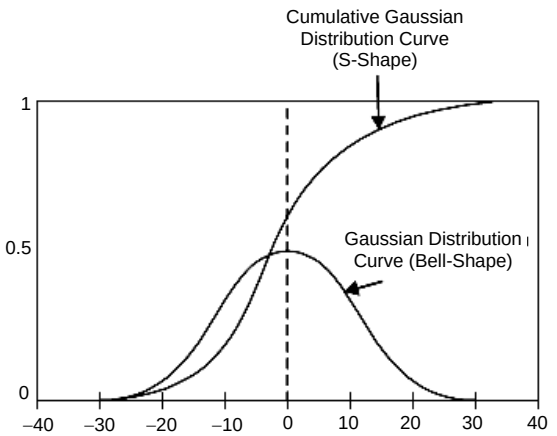
$$= \left(\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right)^2$$

$$= \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}$$

Option (D) is correct.

3. (D)

Gaussian distribution also known as normal distribution is a bell-shaped curve. The cumulative Gaussian distribution curve is S-shaped curve.



4. 0.5 to 0.5

$$\lim_{x \to 1} \frac{1}{\sin x} - \frac{1}{x-1} \qquad (\infty - \infty \text{ form})$$

$$\lim_{x \to 1} \frac{1 - x - \ln x}{\ln x (x-1)} \qquad \left(\frac{0}{0} \text{ form} \right)$$

Apply L-Hospital Rule,

$$\lim_{x \to 0} \frac{1 - \frac{1}{x}}{\frac{1}{x}(x-1) + \ln x}$$

$$\lim_{x\to 0}\frac{1-\frac{1}{x}}{\frac{1}{x}(x-1)+\ln x} \qquad \left(\frac{0}{0}\text{ form}\right)$$

Again apply L-Hospital Rule,

$$\lim_{x\to 1}\frac{\frac{1}{x^2}}{\frac{1}{x^2}+\frac{1}{x}} = \frac{1}{2}$$

5. 15 to 15

$$\text{Volume} = \iiint_V 8xyz\,dy \qquad = 8\int_0^1\int_1^2\int_2^3 xyz\,dx\,dy\,dz$$

$$= 8\int_0^1 z\,dz\int_1^2 y\,dy\int_2^3 x\,dx \qquad = 8\left(\frac{z^2}{2}\right)_0^1\left(\frac{y^2}{2}\right)_1^2\left(\frac{x^2}{2}\right)_2^3$$

$= 1 \times 3 \times 5$

$= 15$

6. (A)

$$\frac{d^2y}{dx^2}+2\frac{dy}{dx}+y=0$$

$(D^2 + 2D + 1)y = 0$

$y(0) = 1,\ y(1) = 3$

Auxiliary equation is,

$m^2 + 2m + 1 = 0$

$m = -1, -1$

$CF = (C_1 + C_2x)e^{-x}$

and PI = 0

Using boundary conditions,

$y(0) = 1,\ C_1 = 1$

$y(1) = 3,\ C_2 = 3e - 1$

So the solution of equation is,

$y = CF + PI$

$y = (C_1 + C_2x)\,e^{-x} + 1$

$y = [1 + (3e - 1)x]e^{-x}$

$y = e^{-x} + (3e - 1)\,xe^{-x}$

7. (B)

Given n = 4, so $h = \frac{b-a}{h} = \frac{1-0}{4} = \frac{1}{4}$

$x_0 = 0 \qquad y_0 = 1$

$x_1 = \frac{1}{4} \qquad y_1 = e^{1/4}$

$x_2 = \frac{1}{2} \qquad y_2 = e^{1/2}$

$x_3 = \frac{3}{4}$ $\qquad$ $y_3 = e^{3/4}$

$x_4 = 1$ $\qquad$ $y_4 = e$

Numerical integration,

$$= \frac{h}{2}[y_0 + y_4 + 2(y_1 + y_2 + y_3)]$$

$$= \frac{1}{8}[(1 + e) + 2(e^{1/4} + e^{1/2} + e^{3/4})]$$

$$= \frac{1}{8}[3.72 + 10.099]$$

$$= 1.727375$$

Option (B) is correct.

8. (A)

$$\lim_{x\to\infty} \frac{x\, ln(x)}{1+x^2} \qquad \left[\frac{\infty}{\infty}\text{form}\right]$$

$$= \lim_{x\to\infty} \frac{1\times ln\, x + x\times(1/x)}{0+2x} \qquad \left[\text{again}\frac{\infty}{\infty}\text{form}\right]$$

$$= \lim_{x\to\infty} \frac{0+(1/x)}{2} \qquad = \lim_{x\to\infty}\frac{1}{2x}$$

$$= \frac{1}{2\times\infty} \qquad = 0$$

9. (C)

$$A = \begin{bmatrix} 5 & 0 & -5 & 0 \\ 0 & 2 & 0 & 1 \\ -5 & 0 & 5 & 0 \\ 0 & 1 & 0 & 2 \end{bmatrix}$$

$$\sim \begin{bmatrix} 5 & 0 & -5 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \end{bmatrix} \quad (R_3 \to R_1 + R_3)$$

$$\sim \begin{bmatrix} 5 & 0 & -5 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1.5 \end{bmatrix} \quad \left(R_4 \to R_4 - \frac{R_2}{2}\right)$$

which is the form of Echelon matrix.

So, the rank of matrix is the number of non-zero rows,

i.e., $\rho(A) = 3$

10. (B)

$$\nabla\phi = 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$$

$$\nabla\phi_{(4,\,4,\,4)} = (2\times 4)\,\hat{i} + (2\times 4)\,\hat{j} + (2\times 4)\hat{k}$$

$$= 8x\hat{i} + 8y\hat{j} + 8z\hat{k}$$

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{8x\hat{i} + 8y\hat{j} + 8z\hat{k}}{\sqrt{8^2 + 8^2 + 8^2}}$$

$$= \frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$$

Option (B) is correct.

11. (A)

A square matrix A is called an orthogonal matrix if the product of matrix A with its transpose matrix A^T is an identity matrix, i.e., $AA^T = I$.

12. 0.0 to 0.0

$$\int_{-1}^{1} x e^{|x|}\,dx = \int_{-1}^{0} x e^{-x}\,dx + \int_{0}^{1} x e^{x}\,dx$$

$$= \left[x(-e^{-x}) - \int(-e^{-x})dx\right]_{-1}^{0} + \left[x(e^{x}) - \int e^{x}\,dx\right]_{0}^{1}$$

$$= \left[-e^{-x}(x+1)\right]_{-1}^{0} + \left[e^{x}(x-1)\right]_{0}^{1}$$

$$= [-e^0 (0 + 1) - (-e (-1 + 1))] + [e(1 - 1) - e^0(0 - 1)]$$

$$= -1 + 1$$

$$= 0$$

Note :

1. A function f is even, if for all values of x

   $f(-x) = f(x)$

   then $\int_{-a}^{a} f(x)\,dx = 2\times\int_{0}^{a} f(x)\,dx$

2. A function f is odd, if for all values of x

   $f(-x) = -f(x)$

   then $\int_{-a}^{a} f(x)\,dx = 0$

Here, $f(x) = x\,e^{|x|}$

$f(-x) = -xe^{|-x|} = -xe^{|x|} = -f(x)$

As the given function is odd.

$$\therefore \int_{-1}^{1} f(x)\,dx = \int_{-1}^{1} x e^{|x|}\,dx = 0$$

13. (A)

$$\frac{dy}{dx} - \frac{y}{x} = 1$$

$$\Rightarrow \frac{dy}{dx} - \underbrace{\left(\frac{-1}{x}\right)}_{P} y = \frac{1}{Q} \qquad \ldots(1)$$

$\therefore$ Integrating factor (IF) $= e^{\int P\,dx} = e^{\int -\frac{1}{x}dx}$

$= e^{-1x\,ln\,x} = x^{-1}$

$= \frac{1}{x}$

Solution of (1) is,

$$y \times IF = \int Q(IF)\,dx + C$$

$$\Rightarrow y \times \frac{1}{x} = \int 1 \times \frac{1}{x}dx + C$$

$$\Rightarrow y \times \frac{1}{x} = ln\,x + ln\,k = ln\,(kx)$$

$$\Rightarrow y = x\,ln\,(kx)$$

14. (A)

$|A - \lambda I| = 0$

$\Rightarrow (2 - \lambda)(6 - \lambda) - 2 = 0$

$\Rightarrow 12 - 2\lambda - 6\lambda + \lambda^2 - 2 = 0$

$\lambda^2 - 8\lambda + 10 = 0$

$\lambda = 4 \pm \sqrt{6}$

Smallest eigen value $= 4 - \sqrt{6} = 1.55$

$$\begin{bmatrix} 2-\lambda & -2 \\ -1 & 6-\lambda \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2-(4-\sqrt{6}) & -2 \\ -1 & 6-(4-\sqrt{6}) \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -2+\sqrt{6} & -2 \\ -1 & 6+\sqrt{6} \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-2x_1 + \sqrt{6}x_1 + 2x_2 = 0$$

$$-2x_1 + \sqrt{6}x_1 = 2x_2$$

$$x_1 = \frac{2x_2}{-2+\sqrt{6}}$$

$$\frac{x_1}{2} = \frac{x_2}{0.45}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0.45 \end{bmatrix}$$

Option (A) is correct.

15. 1 to 1

$$f(x, y) = x\,e^{y}$$

$$\nabla\phi = \Delta f$$

$$= \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k} \qquad = e^{y}\,\hat{i} + x e^{y}\,\hat{j} + 0\hat{k}$$

$$\nabla\phi_{(2,\,0)} = \hat{i} + 2\hat{j}$$

$$\vec{a} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} \qquad = -\frac{3}{2}\hat{i} + 2\hat{j}$$

$$\hat{n} = \frac{\vec{a}}{|\vec{a}|} = \frac{-\frac{3}{2}\hat{i} + 2\hat{j}}{\sqrt{\left(-\frac{3}{2}\right)^2 + (22)^2}} = \frac{2}{5}\left(-\frac{3}{2}\hat{i} + 2\hat{j}\right)$$

$$= -\frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}$$

$$DD = \nabla f\ \hat{n}$$

$$= -\frac{3}{5} + 2 \times \frac{4}{5} \qquad = 1$$

16. 0.38 to 0.42

a = 0, b = 0.5, n = 5

$$h = \frac{b-a}{n} = \frac{0.5-0}{0.5} = 0.1$$

$x_0 = 0,\quad y_0 = 0$
$x_1 = 0.1\quad y_1 = 0.8$
$x_2 = 0.2\quad y_2 = 1.2$
$x_3 = 0.3\quad y_3 = 1.2$
$x_4 = 0.4\quad y_4 = 0.8$
$x_5 = 0.5\quad y_5 = 0$

Numerical integration,

$$= \frac{h}{2}[(y_0 + y_5) + 2(y_1 + y_2 + y_3 + y_4)]$$

$$= \frac{0.1}{2}[(0 + 0) + 2(0.8 + 1.2 + 1.2 + 08)]$$

$$= \frac{0.1 \times 8}{2}$$

$$= 0.4$$

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